

Privacy Regulation and Optimal Product Design

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Abstract

This paper studies how privacy regulation shapes a seller's product design. In a circular city environment, the firm chooses a product design before observing a noisy signal of a consumer's preferences and setting a personalized price. Privacy regulation determines the informativeness of this signal and limits the firm's ability to target consumers. I show that the seller's optimal product design responds non-monotonically to privacy. Greater privacy can induce either broader or more niche products. The extent depends on how it interacts with the firm's pricing incentives. I then examine the resulting implications for consumer welfare.

JEL Classification: D82; L15; L11

1 Introduction

Firms increasingly use consumer data to personalize offerings, especially in pricing and product design. While this personalization can benefit consumers by better matching their preferences, it also raises concerns about privacy and the extent to which firms should be allowed to track individuals. For example, platforms such as Amazon commit to a product catalog and design in advance, but dynamically adjust prices as consumers browse. The platform collects signals about each shopper's preferences (past purchases, search queries, click patterns, etc.) which it uses to tailor prices in real time. The precision of these signals depends on how much data the platform can observe, which is directly affected by privacy regulations. Stronger privacy regulations limit the availability and usage of consumer data, resulting in less informative signals. Weaker privacy protections enable more precise targeting by the platform. Recent initiatives such as the EU's General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) highlight the growing policy focus on how consumer data is used and protected.

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While much of the debate centers on consumer welfare, firms' strategic responses to privacy regulation are equally important. Understanding firm responses matters for policy design: if regulators focus solely on consumer-side protections without accounting for how firms may adapt, they may generate unintended market consequences. In this paper, I examine how limiting a seller's access to consumer information through privacy regulation shapes product design and market outcomes. Existing research has primarily analyzed how privacy affects pricing decisions, but far less is known about its impact on product design, which is a key determinant of which consumers are served and how well their preferences are matched. How should we expect a firm to redesign its product when it knows less about its consumers? One natural intuition is that less information should push the firm toward a broader, less targeted design intended to appeal to the average consumer. But this intuition is not obviously correct. Privacy regulation reshapes the firm's trade-off between consumer reach and match quality, and which force dominates as information becomes more restricted is ambiguous. A firm might broaden its product design to hedge against uncertainty, or it might choose a more niche design tailored closely to a narrower segment. The focus of this paper is to see how product design changes as the degree of privacy changes.

To analyze how privacy regulation shapes product design, I build on the tractable framework of [Bar-Isaac et al. \(2023\)](#), who combine horizontal and vertical differentiation in the spirit of [Salop \(1979\)](#) and [Mussa and Rosen \(1978\)](#). Consumers are located on the circumference of a circle, and a seller chooses a price and a product design. Product design is modeled as a two-dimensional decision combining horizontal (taste variation) and vertical (specialization) differentiation. In this context, product design ranges from broad to niche. This trade-off gives rise to a demand-rotation mechanism in the sense of [Johnson and Myatt \(2006\)](#), where changes in design alter the slope of demand. A broader design increases appeal to marginal consumers (flattening demand), but may reduce the utility of those who strongly preferred the original, more tailored design. Conversely, a niche design provides a better match for a specific segment of consumers, but excludes all other consumers, causing demand to become steeper. Thus, the seller faces a trade-off between consumer reach and match quality when choosing how broad or niche its product should be. I extend this line of research by introducing privacy regulation as a new determinant of product design.

I provide a micro-foundation for incorporating privacy regulation into a model of product design. In this framework, product design depends on the firm's knowledge of consumers' preferences, which is determined by the precision of a signal the seller receives about each consumer's location. If there is no privacy regulation, the signal is fully informative and the seller can learn the consumer's exact preferences. In contrast, if there exists full privacy, the signal is uninformative and reveals nothing about the consumer's tastes. Any intermediate level of privacy regulation allows the seller to learn partially about the consumer. This can also be interpreted as variation in the firm's targeting technology, with more precise technologies corresponding to less privacy. I

show how the level of information available to a seller affects its product design and pricing decisions. Beyond firm behavior, I also study the welfare implications of privacy regulation. Since the seller's product design and pricing decisions shape both who is served and how well their preferences are matched, the effect of privacy regulation on consumer surplus is not obvious. I compare the resulting outcomes to both the first-best and second-best benchmarks.

I find that the seller's optimal product design decision is non-monotonic in privacy regulation. When privacy is initially introduced, the seller responds by broadening its product design, choosing a less specialized offering that appeals to a wider set of consumers. The reason is that less precise information weakens the seller's ability to extract surplus from any given consumer, so the seller compensates by expanding consumer reach through a broader design. However, beyond some threshold of privacy regulation, the seller reverses course and optimally chooses a more niche product design. Once the signals become sufficiently noisy, the seller can no longer charge marginal consumers prices that reflect their willingness to pay, so expanding consumer reach is no longer profitable. The seller instead concentrates on extracting surplus from a smaller group of consumers for whom its product provides a high match value. The seller therefore faces the same trade-off between consumer reach and match quality at all levels of privacy regulation. The main difference is the relative profitability of these two margins, generating this pattern in the seller's optimal behavior.

These product design responses have important welfare implications. Since the seller's design choice shifts non-monotonically with privacy, consumer surplus does as well. Stronger privacy regulation does not uniformly benefit consumers, and in some regions of privacy it can leave certain consumers worse off than weaker regulation would. I compare the profit-maximizing product design and resulting welfare outcome with the first-best and second-best benchmarks, quantifying the welfare costs induced by the seller's product design and pricing decisions. These results highlight a novel channel through which privacy regulation shapes market outcomes: by endogenously altering firms' product design incentives, rather than merely constraining prices or information use.

This paper is most closely related to the broad literature highlighting the trade-offs between consumer privacy and firms' strategic behavior. One strand studies how privacy regulation reshapes market outcomes. [Campbell et al. \(2015\)](#) show that an opt-in regime, in which consumers incur a one-time cost to consent to data use, can advantage large firms and incumbents, reducing entry and raising concentration. [Shy and Stenbacka \(2016\)](#) find that the relationship between privacy protection and industry profits is non-monotonic, and characterize when stronger privacy raises consumer surplus and total welfare. [Lefouili et al. \(2024\)](#) study a monopolist's innovation incentives under a privacy regulation that caps the disclosure of consumer data. Although consumers choose whether to share information, the cap effectively acts as an exogenous policy constraint.

My paper builds on these insights but changes the margin of analysis. This literature asks how

privacy regulation affects market outcomes (e.g. competition, profits, welfare, innovation), taking the firm’s product as given. I ask instead how privacy shapes the product itself. Limits on information precision constrain the seller’s ability to personalize its offering, governing the choice between a broad and a niche design. In modeling privacy as a constraint on information precision, I build on [Bergemann and Bonatti \(2011, 2015\)](#), who characterize how the precision of targeting technologies determines the value of information and the resulting market segmentation in advertising markets. Their framework provides the conceptual foundation for interpreting privacy as a constraint on targeting accuracy.

Complementing this theoretical perspective, a growing empirical literature examines the effects of privacy regulation on firm behavior and market outcomes. [Goldfarb and Tucker \(2011\)](#) document how EU privacy regulation reduced the effectiveness of online advertising, while [Johnson et al. \(2023\)](#) study how GDPR affected market concentration among web technology vendors. [Aridor et al. \(2023\)](#) consider how GDPR changed firms’ ability to identify consumers and engage in targeted advertising. While these empirical studies focus primarily on advertising effectiveness, my paper highlights a distinct market outcome, product design, extending the privacy literature into a dimension that has received limited attention.

For broader discussions of the economics of privacy, see [Acquisti et al. \(2016\)](#), which provides a comprehensive overview of both theoretical and empirical approaches to privacy. [Goldfarb and Que \(2023\)](#) focuses on the role of privacy in digital data markets, highlighting how data collection and usage shape firm behavior and market outcomes. Finally, [Goldfarb and Tucker \(2024\)](#) is a book that offers a systematic treatment of the economics of privacy across a range of institutional settings, including detailed discussions of GDPR and privacy regulation in sectors such as healthcare.

The remainder of this paper is organized as follows. [Section 2](#) introduces the model. [Section 3](#) analyzes two benchmark privacy regimes. [Section 4](#) characterizes the seller’s personalized pricing problem, while [Section 5](#) studies the seller’s optimal product design choice. [Section 5.3](#) examines welfare outcomes, including the social planner’s problem. [Section 6](#) concludes with policy implications and discusses directions for future research.

2 Model

2.1 Players

There is a unit mass of consumers uniformly distributed along the circumference of a circle of radius one. Each consumer (she) is identified by a location $\theta_c \in [0, 2\pi)$, which represents her taste parameter. The seller (he) is located at $\theta_S = 0$, without loss of generality, and offers a product

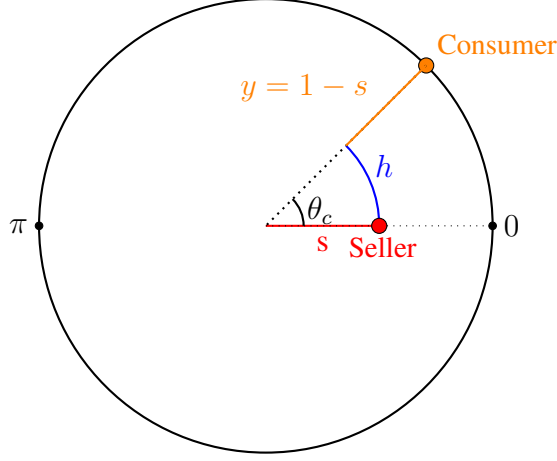


Figure 1: Consumer mismatch costs under a given product design s . A consumer located at θ_c first incurs a vertical cost $c_v(1 - s)$ by moving radially inward a distance $y = 1 - s$; the corresponding segment is drawn in orange. She then travels along the circular arc at radius s (in blue) to the seller's location $\theta_s = 0$, incurring a horizontal cost $c_h = th$, where $h = s \cdot d_0(\theta_c, 0)$ and $d_0(\theta_c, 0)$ is the geodesic distance on the circle.

design $s \in [s_{\min}, 1]$ at price p , produced at constant marginal cost mc . This design parameter s determines the seller's position along that radius: smaller values of s place the firm closer to the center (a broader design), while larger values place it closer to the circumference (a more niche design).

Consumption involves two stages of mismatch relative to the seller's offered product, illustrated in Figure 1. When the seller chooses a product design s , consumers must first adjust by an amount $1 - s$, incurring a vertical cost $c_v(1 - s)$, where $c'_v(\cdot) > 0$ and $c''_v(\cdot) > 0$. This reflects the degree of specialization of the product: a higher s corresponds to a more specialized design, reducing the vertical transportation cost for all consumers. Given s , this cost is identical for all consumers. Consumers also differ in their idiosyncratic tastes. A consumer located at θ_c faces a horizontal transportation cost proportional to the geodesic distance $d_0(\theta_c, 0)$. The corresponding cost is $c_h(s) = tsd_0(\theta_c, 0)$, where $t > 0$ measures the intensity of the horizontal differentiation and s scales how localized preferences are within the circle. Thus, a consumer's transportation costs represent her disutility from preference misalignment, combining a vertical loss from a less tailored quality and a horizontal loss from differing tastes. These two costs captures the seller's tradeoff: choosing a more niche product design can reduce the vertical cost for all consumers, but can increase the horizontal cost for some consumers.

Combining these components, the utility from purchasing is

$$U(\theta_c; s, p) = V - c_v(1 - s) - c_h(\theta_c; s) - p,$$

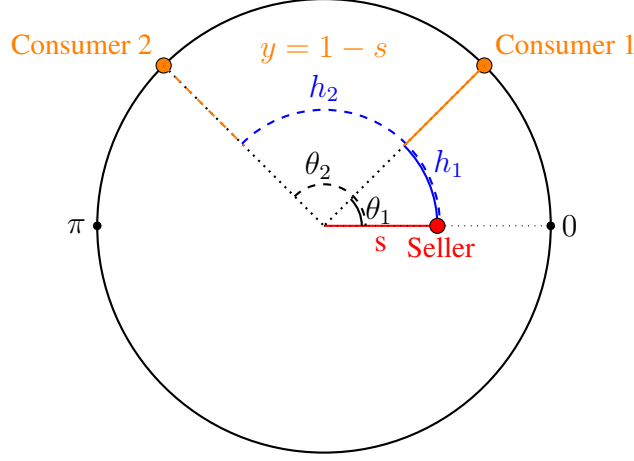


Figure 2: Two consumers located at θ_1 and θ_2 face a product design s . The vertical distance $y = 1 - s$ is identical for both, but the horizontal distance $c_h(s)$ varies. In particular, consumer 2 faces a higher horizontal transportation cost, since $h_2 > h_1$. Raising s therefore reduces vertical mismatch uniformly while increasing horizontal mismatch for every consumer in proportion to her location, the seller's fundamental tradeoff in choosing product design.

where V is the baseline product valuation and p is the price charged. A consumer buys the product if $U(\theta_c; s, p) \geq 0$.

The seller's objective in choosing product design s and price p is to maximize profit. However, choosing design s entails a fundamental trade-off. Consider two consumers located at θ_1 and θ_2 , depicted in Figure 2. When the seller selects a product design s , both consumers face the same vertical transportation cost, but different horizontal transportation costs since $h_2 > h_1$. If the seller chooses a more niche product design, both consumers face a lower vertical transportation cost and higher horizontal transportation cost, as h_1 and h_2 increase. However, while this change can decrease consumer 1's total distance, h_2 increases substantially, inducing a much higher horizontal cost for consumer 2. In contrast, if the seller selects a broader product design, consumer 2 benefits from a lower horizontal cost, but consumer 1 can be worse off, incurring a higher vertical cost. Therefore, when deciding between a broader or more niche product design, the seller has to consider both the vertical and horizontal transportation of consumers, and depending on where the consumer is located, this tradeoff changes. This fundamental trade-off in product design becomes particularly consequential when I introduce privacy regulation, which limits the information the seller can use to balance it.

2.2 Privacy

The seller learns about the consumer's preferences through a noisy signal x . The prior belief is that consumers are uniformly distributed along the circumference of the circle. Given a consumer's

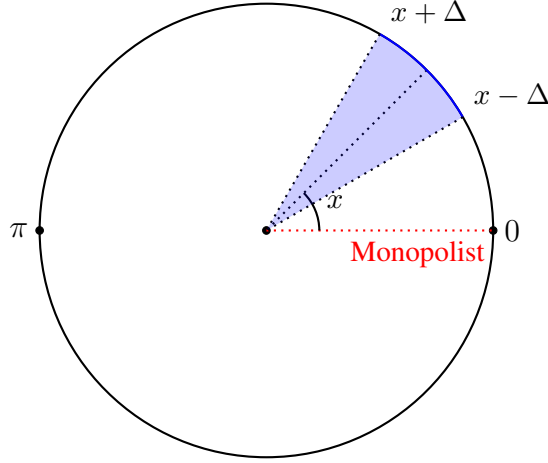


Figure 3: Let Δ represent the level of privacy regulation. Upon receiving a noisy signal x , the monopolist's posterior is that the consumer is distributed uniformly over $[x - \Delta, x + \Delta]$.

taste parameter θ_c , the signal realization x is drawn from a uniform distribution,

$$x \mid \theta_c \sim U[\theta_c - \Delta, \theta_c + \Delta].$$

The signal x is centered around θ_c and where $\Delta \geq 0$ represents the level of privacy regulation, with higher Δ reflecting stronger privacy. At the upper limit $\Delta = \pi$, the signal window spans the entire circle; at $\Delta = 0$, the signal reveals θ_c exactly.

Upon observing x , the seller Bayesian updates his belief about the consumer's preference: the posterior is uniform over the geodesic preference arc of radius Δ around x ,

$$S(x, \Delta) = \{\theta_c : d_0(\theta_c, x) \leq \Delta\}.$$

This posterior interval is illustrated in Figure 3. Larger values of Δ correspond to noisier signals and therefore stronger privacy regulation, reducing the precision of the seller's information about the consumer. This posterior belief determines the seller's personalized pricing problem.

2.3 Timing

The model timing is as follows. The level of privacy, Δ , is fixed and public knowledge. Given Δ , the seller sets a product design s that is offered to all consumers. Nature then selects a consumer and generates a signal x about that consumer's preferences, following the signal generating process discussed in Section 2.2. The seller observes the noisy signal x of a consumer's preferences and sets a personalized price, based on his posterior $S(x, \Delta)$.

This timing implies that firms can adjust prices flexibly while product design remains fixed. In practice, this seems plausible, as noted in [Tirole \(1988\)](#), “Prices can often be adjusted faster than product characteristics can.” I solve for the subgame perfect Nash equilibrium (SPNE).

It is worth noting that if the seller could choose both product design and price after observing x , the problem would be simpler. In that case, design would be tailored to the specific posterior interval $S(x, \Delta)$ rather than the entire circle. Privacy would still matter, since a larger Δ implies a broader segment to serve, but the effect would be monotonic: greater privacy would always push the seller toward a broader design. Requiring commitment to s before observing x forces the seller to balance vertical and horizontal mismatch across all possible consumer tastes. Thus, I focus on the ex-ante design choice, which captures how privacy regulation affects product design in a more meaningful way.

This structure mirrors settings such as online retail platforms, where firms commit to product design in advance but adjust prices in real time using consumer data. The model therefore captures how the informativeness of consumer signals, which is shaped by privacy regulation, affects a seller’s design and resulting market outcomes.

3 Benchmarks

I begin by analyzing two benchmark cases: no information and perfect information. These correspond to the extremes of the privacy parameter Δ . No information arises when $\Delta = \pi$, producing a completely uninformative signal, under which the seller’s posterior coincides with the prior. Perfect information arises when $\Delta = 0$, yielding a fully informative signal, perfectly revealing consumer’s preferences. This is the case of no privacy. I focus on signals on the half-circle, $x \in [0, \pi]$ due to the symmetric nature of the full circle.

3.1 No Information: $\Delta = \pi$

Consider the first extreme where the seller has no information about the consumer’s location. This corresponds to the maximum privacy regulation: the signal x reveals nothing, and the seller’s posterior belief about consumer location remains $U[0, \pi]$, corresponding to the maximal geodesic half-width $\Delta = \pi$.

Under this case, the seller cannot personalize prices and must set a uniform price $p(s)$. Given a product design s and price p , a consumer located at θ_c purchases the product if

$$V - c_v(1 - s) - ts\theta_c - p(s) \geq 0 \Rightarrow \theta_c < \frac{V - c_v(1 - s) - p(s)}{ts} = \hat{\theta}_c. \quad (1)$$

Intuitively, a consumer purchases when their net surplus is non-negative. Holding design and price fixed, only consumers sufficiently close to the seller meet this condition.

The resulting consumer demand is given by

$$Q(s, p(s)) = \min \left\{ 1, \frac{V - c_v(1 - s) - p(s)}{ts\pi} \right\},$$

where $Q(s, p(s)) = 1$ when there is full market coverage. Otherwise, there is partial market coverage, where the demand corresponds to the probability that (1) holds.

The seller's pricing problem is therefore to choose the optimal price that maximizes their profit:

$$\max_{p(s)} \Pi(s, p(s)) = (p(s) - mc) \cdot Q(s, p(s)).$$

The optimal uniform price is therefore

$$p^*(s) = \begin{cases} V - c_v(1 - s) - ts\pi & \text{if } t \leq \frac{V - c_v(1 - s) - mc}{2s\pi}, \\ \frac{V - c_v(1 - s) + mc}{2} & \text{if } t > \frac{V - c_v(1 - s) - mc}{2s\pi}. \end{cases}$$

When the horizontal transportation cost t is sufficiently large, consumers located farther away may have willingness-to-pay below the uniform price, resulting in partial market coverage. Intuitively, full market coverage corresponds to the case where even the farthest consumer's willingness-to-pay exceeds the seller's profit-maximizing price, so the seller finds it profitable to set a uniform price low enough to serve the entire market. Partial coverage, by contrast, arises when horizontal transportation costs are high enough that serving distant consumers would require a price cut too deep to justify. The seller instead extracts higher margins from a smaller served consumer base. This markup-coverage tradeoff is a direct consequence of the uniform-pricing constraint imposed by the no-information benchmark. Since the seller is unable to distinguish consumers by location, he must set the same price for all consumers, rather than adjusting prices based on the signal he receives.

Optimal Product Design. The seller chooses the product design that maximizes profit, thus, his maximization problem is as follows:

$$\max_s \Pi(s) = (p^*(s) - mc)Q(s, p^*(s)).$$

Solving this problem yields the following first-order conditions, which I interpret below. Under full market coverage, the seller's optimal product design solves the following first order condition:

$$c'_v(1 - s) = t\pi. \quad (2)$$

In the partial market coverage case, the optimal product design choice satisfies the following condition:

$$c'_v(1 - s) = t\pi Q(s). \quad (3)$$

In both cases, the seller's optimal product design balances two channels through which s affects consumers' willingness-to-pay. The LHS, $c'_v(1 - s)$, captures the marginal vertical channel. A higher s corresponds with a more niche design, which reduces the consumer's vertical transportation cost at rate $c'_v(1 - s)$, raising willingness-to-pay uniformly across all consumers. The RHS captures the marginal horizontal channel, which is heterogeneous across consumers. A more niche product leaves consumers at $\theta_c = 0$ unaffected horizontally, but consumers farther around the circle experience increasing horizontal disutility. Raising s therefore makes the product a better match for consumers near the seller and worse matched for distant consumers. Since the seller cannot condition s on individual consumer locations, he must select a product design that balances this heterogeneous response. Under full market coverage, the binding constraint comes from the farthest consumer at $\theta_c = \pi$, whose marginal horizontal disutility $t\pi$ pins down the uniform price. Under partial market coverage, the horizontal margin reflects only the served fraction $Q(s)$ of the market.

Lemma 1 *Under no information, where $\Delta = \pi$, the seller's optimal product design satisfies*

(i) $c'_v(1 - s) = t\pi$ *under full market coverage, and*

(ii) $c'_v(1 - s) = t\pi Q(s)$ *under partial market coverage,*

where $Q(s) = \frac{V - c_v(1-s) - mc}{2ts\pi}$ represents the fraction of consumers served in the partial market coverage case.

3.2 Perfect Information: $\Delta = 0$

When $\Delta = 0$, the posterior support collapses to a single point: the seller observes the consumer's exact taste parameter $x = \theta_c$. Given design s and price p , the consumer located at $x = \theta_c$ buys if their valuation is positive:

$$V - c_v(1 - s) - tsx - p(s, x) \geq 0. \quad (4)$$

Since the seller has perfect information about the consumer's preferences, he sets a price equal to the consumer's willingness-to-pay when doing so is profitable, and excludes consumers otherwise. The seller's optimal price is therefore

$$p^*(s, x) = \max\{V - c_v(1 - s) - tsx, mc\}.$$

Given this price, the marginal consumer is given by

$$\hat{\theta}_c = \frac{V - c_v(1 - s) - mc}{ts}.$$

Under partial market coverage, only consumers $\theta_c \leq \hat{\theta}_c$ are served at personalized prices, while consumers further away are excluded, since the seller's optimal price would fall below marginal cost. Demand is therefore given by

$$Q(s) = \min \left\{ 1, \frac{V - c_v(1 - s) - mc}{ts\pi} \right\}.$$

Optimal product design. The seller chooses an optimal product design that maximizes his expected profit, since product design is an ex-ante choice. In the full market coverage case, in which all consumers are profitably served and prices equal each consumer's willingness-to-pay, the seller's expected profit is

$$\Pi(s) = V - c_v(1 - s) - mc - \frac{ts\pi}{2}.$$

Otherwise, under the partial coverage case, in which some consumers have willingness-to-pay below marginal cost and are therefore optimally excluded (equivalently, pricing at $p = mc$ does not induce purchase), the seller's expected profit under partial coverage is

$$\Pi(s) = \frac{(V - c_v(1 - s) - mc)^2}{2ts\pi}.$$

Under full market coverage, the seller sets a personalized price that even the farthest consumer at $x = \pi$ is willing to pay. Thus, the full-coverage design under perfect information satisfies the following first order condition:

$$c'_v(1 - s) = \frac{t\pi}{2}. \tag{5}$$

Under partial market coverage, some consumers have a willingness-to-pay below the marginal cost

m , and thus are excluded. The optimal product design choice then satisfies the following condition:

$$c'_v(1 - s) = \frac{t\pi}{2}Q(s). \quad (6)$$

As it was under no information, the seller's optimal product design balances the same two channels: vertical distance on the LHS and horizontal distance on the RHS. The LHS, $c'_v(1 - s)$, again captures how a more niche product design reduces vertical transportation cost uniformly across all consumers, raising their willingness-to-pay. The RHS captures the marginal horizontal channel, but the scaling differs from the no information case. Under perfect information, the seller observes each consumer's location, and therefore can price each individually at her willingness-to-pay. The horizontal margin therefore reflects the average increase in horizontal transportation cost across served consumers. Under full market coverage, the seller serves all consumers over $[0, \pi]$, so the average marginal horizontal disutility of increasing s is $\frac{t\pi}{2}$. Under partial market coverage, the average is taken over the served consumers, denoted by the demand function $Q(s)$. The factor of $\frac{1}{2}$ relative to the no-information case has a clean economic interpretation. Without information, the seller is constrained to a uniform price, and has to compensate the marginal consumer for her horizontal disutility. With perfect information, however, the seller engages in personalized pricing, so thus focuses on the average disutility across served consumers. Due to consumers being distributed uniformly, the average is half the maximum.

Lemma 2 *Under perfect information, where $\Delta = 0$, the seller's optimal product design satisfies*

- (i) $c'_v(1 - s) = \frac{t\pi}{2}$ under full market coverage, and
- (ii) $c'_v(1 - s) = \frac{t\pi}{2}Q(s)$ under partial market coverage,

where $Q(s) = \frac{V - c_v(1-s) - mc}{ts\pi}$ represents the fraction of consumers served in the partial market coverage case.

3.3 Full vs. No Information Comparison

The two benchmarks highlight how information shapes the seller's product design and pricing decisions. Under no information, the seller is constrained to a uniform price. Under perfect information, however, the seller can fully personalize prices to extract all surplus up to the marginal buyer. This allows the seller to partially offset the higher horizontal mismatch faced by distant consumers as design becomes more niche. Under no information, the uniform price cannot compensate these consumers, so a broader design is optimal.

Proposition 1 *The seller's optimal product design under perfect information is weakly more niche than under no information:*

$$s_P \geq s_N.$$

The value of information to the seller depends on market coverage. Under full market coverage, the seller is constrained by the farthest consumer's participation. If $\Delta = \pi$, this constraint binds at the farthest consumer ($x = \pi$), whose marginal horizontal disutility is $t\pi$, while under perfect information, the seller can price individually, and the margin becomes $\frac{t\pi}{2}$. The seller therefore chooses a more niche design under perfect information. Under partial market coverage, by contrast, the seller's design choice is pinned down by the marginal consumer ($x < \pi$) in both information regimes. Information changes which consumers surplus is extracted from, but not optimal design itself.

4 Seller's Optimal Strategy

The benchmark cases provide intuition for how information precision affects firm product design choice. However, they do not provide a complete understanding of the effect of privacy regulation. Thus, I consider the case of intermediate privacy, where $\Delta \in (0, \pi)$, which is the focus of the remaining analysis. I first characterize the seller's optimal pricing rule for a given product design and signal realization, then use these results to study how the ex-ante product design responds to the level of privacy regulation.

4.1 Seller's Problem

Given a design s and price p , a consumer with taste parameter θ_c purchases if

$$V - c_v(1 - s) - c_h(s) - p \geq 0.$$

For a fixed design s , the price p determines how far along the circle consumers are willing to buy. Define the following cutoff distance as the seller's acceptance reach:

$$\theta_r(p; s) = \frac{V - c_v(1 - s) - p}{ts}, \quad \mathcal{R}(\theta_r) = \{\theta_c : d_0(\theta_c, 0) \leq \theta_r(s, p)\}. \quad (7)$$

Economically, θ_r captures the maximum taste mismatch consumers are willing to tolerate. It identifies the marginal consumer who is indifferent between purchasing a product with design s and at price p and not purchasing at all. Consumers whose taste differs more than θ_r choose not to buy. A higher price corresponds to a smaller reach, and vice versa, so the seller can equivalently be viewed

as choosing either a price p or an acceptance reach θ_r . The acceptance region, $\mathcal{R}(\theta_r)$, thus consists of the consumers who would purchase the product. To determine the set of consumers that may actually be in the market, the seller combines this willingness-to-buy region with his information about the consumer's location.

4.1.1 Consumer Preference Set

After observing signal x , the seller updates his belief about the consumer's taste. The posterior support depends on the level of privacy regulation: a noisier signal yields a wider arc of possible consumer locations, while a precise signal narrows it to a small neighborhood. Formally, the posterior support is

$$S(x, \Delta) = \{\theta \in [0, 2\pi) : d_0(\theta, x) \leq \Delta\}.$$

This representation nests the benchmark cases: when $\Delta = 0$, the posterior collapses to the single point $\{\theta_c\}$; when $\Delta = \pi$, the posterior covers the entire circle. Without loss of generality, I restrict to $x \in [0, \pi]$ (as in [Section 3](#)), so that $d_0(x, 0) = x$. Combining the posterior support with the acceptance region gives the set of consumers who are both potential matches and willing to buy at price p .

Formally, the relevant set of buyers, given design s , price p , signal x , and privacy Δ , is

$$B(x, \Delta, \theta_r) = S(x, \Delta) \cap \mathcal{R}(\theta_r).$$

The measure of this set is

$$L(x, \Delta, \theta_r) = |B(x, \Delta, \theta_r)|,$$

which can be characterized by the cases depicted in [Figure 4](#). In particular, the seller either faces full market coverage, partial market coverage, or no coverage, depending on how the posterior and acceptance region overlap. Under full market coverage, the posterior support falls entirely within the acceptance region, depicted in [Figure 4a](#). What limits demand in this case is the width of the posterior (i.e. privacy), rather than the reach. Conditional on x , he believes no consumers lie beyond $x + \Delta$, so extending his reach further would serve no one. Partial market coverage can be divided into two cases. When the acceptance region falls within the posterior support, the served set of consumers is limited by the seller's reach, shown in [Figure 4b](#). [Figure 4c](#) captures the case where some consumers within the seller's posterior support fall inside the acceptance region and would purchase, while others lie beyond the acceptance reach and would not. Crucially, not all consumers in the posterior support are served, unlike in the full market coverage case. Finally, [Figure 4d](#) corresponds to the case of no purchase, where the seller's posterior belief lies entirely

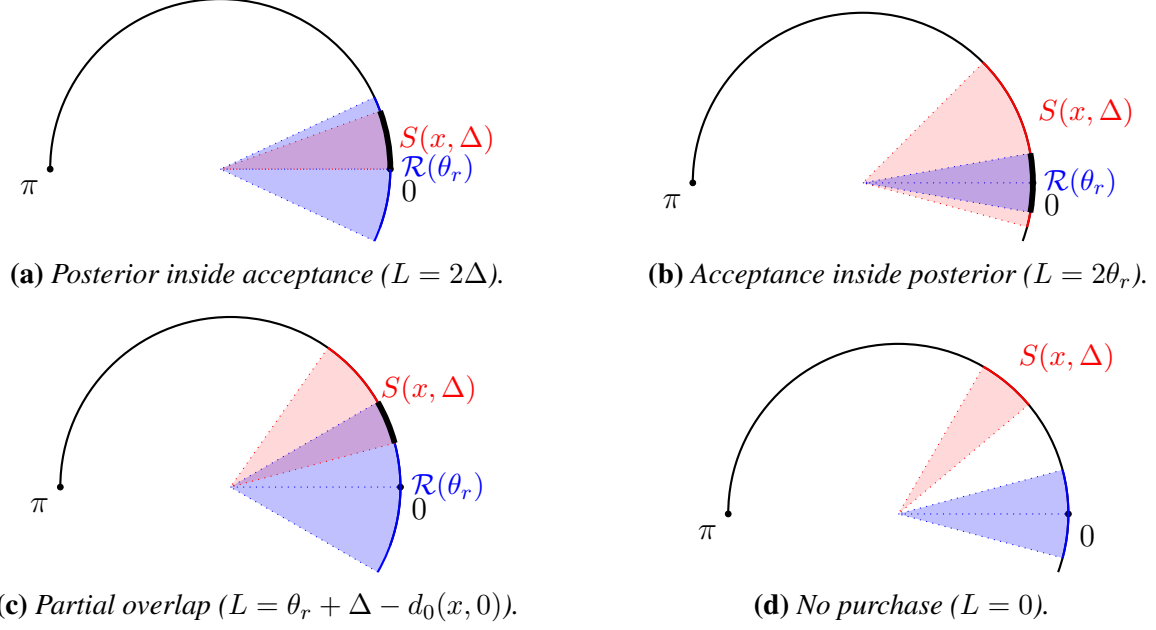


Figure 4: Four geometric cases for the buyer-length function $L(x, \Delta, \theta_r)$ on the half-circle. Red shading shows the posterior support $S(x, \Delta)$ and blue shading shows the acceptance region $\mathcal{R}(\theta_r)$. The overlap area corresponds to the buyer length $L(x, \Delta, \theta_r)$, denoted by the bold black arc.

outside the acceptance region, so $L = 0$.¹

The demand, conditional on privacy level Δ and signal x , is therefore

$$Q(p, s, x, \Delta) = \frac{L(x, \Delta, \theta_r)}{2\Delta}. \quad (8)$$

Q is the share of potential consumers the seller actually serves, with $Q = 1$ under full market coverage and $Q < 1$ under partial market coverage. Privacy shapes demand by determining which regime applies, since Δ governs how much of the posterior support lies within the seller's reach.

4.1.2 Pricing

Given (s, x, Δ) , the seller sets a price to maximize expected profit,

$$\Pi(p | s, x, \Delta) = (p - mc) Q(p, s, x, \Delta),$$

where $Q(p, s, x, \Delta)$ follows (8).

Since price and reach are inversely related, profit can equivalently be expressed as a function of θ_r :

$$\Pi(\theta_r | s, x, \Delta) = \frac{(p(\theta_r) - mc) L(x, \Delta, \theta_r)}{2\Delta}.$$

¹The closed form characterization of $L(x, \Delta, \theta_r)$ can be found in [Appendix A.1](#).

The seller only serves consumers at prices above marginal cost, $p(\theta_r) \geq mc$. Using (7), $p(\theta_r) = V - c_v(1 - s) - ts\theta_r$, which imposes an upper bound on the acceptance reach; the seller's price covers marginal cost only for θ_r small enough that $V - c_v(1 - s) - ts\theta_r \geq mc$.

4.1.3 Optimal Acceptance Reach

Given the profit function $\Pi(\theta_r | s, x, \Delta)$, the seller chooses the acceptance range θ_r (equivalently, the price p) to maximize profit. Intuitively, this determines which subset of consumers the seller serves. Since $L(x, \Delta, \theta_r)$ depends on the relative position of the posterior and the acceptance region, the maximization problem splits into three geometric cases in which trade occurs, depicted in Figure 4a-Figure 4c: posterior inside acceptance (sell-to-all), acceptance inside posterior (interior), and partial overlap (partial). In particular, the sell-to-all regime corresponds to the full market coverage case, while the interior and partial cases are the two forms of partial market coverage. In each trade regime, the seller's problem reduces to a one-dimensional optimization over θ_r . The following lemma collects the measure of buyers, expected profit, and optimal reach for each case; all feasibility conditions and boundary cases are deferred to Appendix A.5.2.

Lemma 3 Fix (s, x, Δ) and let $\bar{V} = V - c_v(1 - s)$ to simplify notation. The three trade cases yield:

(i) **Sell-to-all** ($S \subset \mathcal{R}$, i.e. $\theta_r \geq x + \Delta$): $L = 2\Delta$ and $\Pi(\theta_r) = \bar{V} - mc - ts\theta_r$,

$$\theta_{r,\text{all}}^* = x + \Delta, \quad p_{\text{all}}^* = \bar{V} - ts(x + \Delta), \quad \Pi_{\text{all}}^* = \bar{V} - mc - ts(x + \Delta).$$

(ii) **Interior** ($\mathcal{R} \subset S$, i.e. $\theta_r < \Delta$ with $x \leq \Delta$): $L = 2\theta_r$ and $\Pi(\theta_r) = \frac{(\bar{V} - mc - ts\theta_r)\theta_r}{\Delta}$,

$$\theta_{r,\text{int}}^* = \frac{\bar{V} - mc}{2ts}, \quad p_{\text{int}}^* = \frac{\bar{V} + mc}{2}, \quad \Pi_{\text{int}}^* = \frac{(\bar{V} - mc)^2}{4ts\Delta}.$$

(iii) **Partial overlap**: $L = \theta_r + \Delta - x$ and $\Pi(\theta_r) = \frac{(\bar{V} - mc - ts\theta_r)(\theta_r + \Delta - x)}{2\Delta}$,

$$\theta_{r,\text{par}}^* = \frac{\bar{V} - mc}{2ts} - \frac{\Delta - x}{2}, \quad p_{\text{par}}^* = \bar{V} - ts\theta_{r,\text{par}}^*, \quad \Pi_{\text{par}}^* = \frac{(\bar{V} - mc + ts(\Delta - x))^2}{8ts\Delta}.$$

In the sell-to-all case, the seller chooses the smallest reach that still covers the posterior, $\theta_{r,\text{all}}^* = x + \Delta$. Since profit falls as the reach grows beyond this point, the firm minimizes the acceptance range while still ensuring full coverage. In the interior case, the optimal reach $\theta_{r,\text{int}}^*$ and price are independent of x and Δ . This is because the posterior boundaries are slack at the optimum. The

acceptance region sits strictly in the interior of the posterior, so the edges do not enter the first order condition. The seller thus prices as though it is facing a local demand problem in which the posterior boundaries do not bind. The partial case maximizer is $\theta_{r,\text{int}}^*$ shifted by $-\frac{1}{2}(\Delta - x)$, reflecting the fact that one posterior boundary now constrains the seller's optimization. When $x \leq \Delta$, more of the posterior mass lies near the firm, reducing the optimal acceptance reach relative to the acceptance-inside-posterior case. When $x > \Delta$, the posterior lies further from the seller, increasing the optimal acceptance reach. The effect of design s on $\theta_{r,\text{int}}^*$ and $\theta_{r,\text{par}}^*$ is ambiguous, as a more niche design raises $\bar{V}$, but also the horizontal mismatch coefficient ts .

For each fixed (s, x, Δ) , the measure of buyers induces exactly one of the geometries in [Figure 4](#), and three cases partition the feasible range of $\theta_r \in [0, \theta_{r,\text{feas}}]$, where $\theta_{r,\text{feas}} = \frac{\bar{V}-mc}{ts}$ is the largest reach consistent with $p \geq mc$. The global maximizer $\theta_r^*(s, x, \Delta)$ is therefore one of the interior maximizers above or the feasibility boundary.

Proposition 2 (Per- x optimal reach and price) *Fix (s, x, Δ) . Form the candidate set*

$$\Theta = \{ \theta_{r,\text{all}}^* \text{ if feasible, } \theta_{r,\text{int}}^* \text{ if feasible, } \theta_{r,\text{par}}^* \text{ if feasible, } \theta_{r,\text{feas}} \}.$$

Then

$$\theta_r^*(s, x, \Delta) \in \arg \max_{\theta_r \in \Theta} \Pi(\theta_r | s, x, \Delta), \quad p^*(s, x, \Delta) = p(\theta_r^*(s, x, \Delta)),$$

with candidate profits as in [Lemma 3](#).

Since the profit functions agree at the regime boundaries, no boundary candidates beyond $\theta_{r,\text{feas}}$ are needed. The feasibility candidate covers the case where the constraint binds before any interior optimum is reached. [Proposition 2](#) thus reduces the per-signal pricing to a comparison across at most four candidates.

In particular, the seller's pricing behavior varies with the signal realization. For signals close to the firm's location and low levels of privacy, the seller can profitably serve the entire posterior (full market coverage). Otherwise, for more distant signals or higher privacy levels, it optimally excludes some consumers (partial market coverage), or does not trade at all. This signal-dependent pricing structure is one of the forces shaping the interaction between privacy and product design analyzed in the next section.

4.1.4 Maximized Profit

Given (s, x, Δ) , the seller's optimal acceptance reach $\theta_r^*(s, x, \Delta)$ and corresponding price $p^*(s, x, \Delta)$ yield the per-signal profit

$$\Pi^*(s, x, \Delta) = \Pi(\theta_r^*(s, x, \Delta)). \tag{9}$$

While pricing is personalized for each signal realization x , the ex-ante product design is chosen to maximize expected profit across signals. By rotational symmetry of the circle, $\Pi^*(s, x, \Delta) = \Pi^*(s, 2\pi - x, \Delta)$. Hence it suffices to consider $x \in [0, \pi]$, where the prior density is $1/\pi$ under the uniform distribution, yielding

$$\Pi(s, \Delta) = E[\Pi^*(s, x, \Delta)] = \frac{1}{\pi} \int_0^\pi \Pi^*(s, x, \Delta) dx. \quad (10)$$

This expected profit—the average of the per-signal optima from [Lemma 3](#) and [Proposition 2](#)—serves as the objective for the ex-ante product design choice analyzed in the next section.

5 Effects of Privacy Regulation

The aggregate expected profit in (10) allows for the analysis of how the seller’s optimal product design responds to different levels of privacy regulation. This section first derives the first-order condition for optimal design, then uses comparative statics to establish that this relationship is non-monotonic. [Section 5.2](#) examines the implications for consumer surplus.

5.1 Optimal Product Design

The seller chooses product design s to maximize expected profit:

$$s^*(\Delta) \in \arg \max_{s \in [s_{\min}, 1]} \Pi(s, \Delta).$$

Differentiating (10) with respect to s and applying Leibniz’s rule and the envelope theorem yields the first-order condition²:

$$\frac{\partial}{\partial s} \frac{1}{\pi} \int_0^\pi \Pi^*(s, x, \Delta) dx = \frac{1}{\pi} \int_0^\pi \frac{L(x, \Delta, \theta_r^*(s, x, \Delta))}{2\Delta} (c'_v(1 - s) - t \theta_r^*(s, x, \Delta)) dx = 0. \quad (11)$$

Intuitively, the integrand represents the marginal effect of a change in product design on per-signal profit. A more niche product (higher s) lowers the vertical mismatch costs ($c'_v(1 - s) > 0$), but raises horizontal mismatch costs through the $t \theta_r^*$ term. The weighting by $L/(2\Delta)$ reflects the share of the posterior that is served at the per-signal optimum ([Proposition 2](#)). Averaging this tradeoff across all signals x yields the equilibrium design $s^*(\Delta)$.

²Leibniz’s rule is valid since $\Pi^*(s, x, \Delta)$ is continuous and piecewise differentiable in s . The envelope theorem eliminates the indirect effect through $\theta_r^*(s, x, \Delta)$.

5.1.1 Non-monotonicity in Optimal Product Design

To study the behavior of the seller's optimal product design choice, I apply the implicit function theorem to the first-order condition in (11):

$$\frac{ds^*}{d\Delta} = - \frac{\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta}}{\frac{\partial^2 \Pi(s, \Delta)}{\partial s^2}}. \quad (12)$$

The numerator is the cross-partial derivative of expected profit, capturing how the marginal incentive to adjust product design changes with privacy. The denominator is the second derivative of expected profit with respect to design. Whenever $s^*(\Delta)$ is an interior maximizer, it satisfies the second-order condition $\frac{\partial^2 \Pi}{\partial s^2} \big|_{s^*} < 0$. Under this condition, the sign of (12) is determined by the sign of the cross-partial $\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta}$.

Since privacy can shift the seller's optimal acceptance behavior across three regimes (sell-to-all, acceptance-inside-posterior, and partial-overlap), the cross-partial inherits a regime-dependent structure. The closed-form expressions for the cross-partial in each regime are derived in [Appendix A.2](#). The total effect of privacy on design incentives is an aggregation of regime-specific effects, each weighted by the measure of signals for which that regime applies, as formalized in [Proposition 3](#).

Proposition 3 (Decomposition of cross-partial across regimes) *For any (s, Δ) , the partition of $[0, \pi]$ into $\mathcal{X}_{\text{all}}$, $\mathcal{X}_{\text{int}}$, $\mathcal{X}_{\text{par}}$ (up to measure-zero boundaries) yields*

$$\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta} = \sum_{R \in \{\text{all}, \text{int}, \text{par}\}} \frac{1}{\pi} \int_{\mathcal{X}_R(s, \Delta)} \frac{\partial^2 \Pi^*(s, x, \Delta)}{\partial s \partial \Delta} dx.$$

In the sell-to-all regime, where the seller serves the entire posterior, the cross-partial is unambiguously negative: greater privacy strictly weakens his incentive to specialize. In the acceptance-inside-posterior and partial-overlap regimes, the sign is ambiguous and depends on the relative strength of the vertical mismatch costs and horizontal targeting effects. As privacy increases, the measure of signals falling into each regime changes, shifting which force dominates. The next section analyzes this shift to derive the comparative statics of $s^*(\Delta)$.

5.1.2 Comparative Statics

First, consider the case of no privacy regulation, which provides the baseline for analyzing how privacy affects the seller's optimal design. When $\Delta = 0$, the seller sells to all consumers, leading to a decreasing price schedule as the realized signal x moves farther from the seller's position.

When privacy increases slightly, the seller can respond in two ways: by adjusting only the price, or by changing product design, which in turn alters the optimal price schedule.

If the seller adjusts price but holds product design fixed, the entire price schedule shifts downward for all signals. This is because the posterior has widened, and in the sell-to-all case, must lower the price to include the consumer located at $x + \Delta$.

If instead the seller changes the product design, the price schedule rotates: broadening the design lowers prices for consumers whose tastes are close to the seller's, but raises them for those with more distinct tastes. Making the product design more niche has the opposite effect—raising prices for nearby consumers while lowering them for those farther away in preference space.

In practice, the seller can choose a combination of price and design adjustments to maximize profit. [Proposition 4](#) formalizes the intuition that when privacy increases slightly from $\Delta = 0$, the seller reduces product differentiation. Prices fall uniformly across consumers, consistent with a rotation of the price schedule described above.

Proposition 4 (Comparative statics near $\Delta = 0$) *Suppose $\theta_{r,\text{feas}}(s^*(0)) > 0$, where*

$$\theta_{r,\text{feas}}(s) = \frac{V - c_v(1 - s) - mc}{ts}$$

is the maximum feasible reach. Then

$$\frac{\partial^2 \Pi^*(s^*(0), 0^+)}{\partial s \partial \Delta} = \begin{cases} -\frac{c'_v(1 - s^*(0))}{\pi} & \text{if } \theta_{r,\text{feas}}(s^*(0)) \leq \pi, \\ -t & \text{if } \theta_{r,\text{feas}}(s^*(0)) > \pi, \end{cases}$$

and hence

$$\frac{ds^*}{d\Delta}(0^+) < 0.$$

This result implies that, close to perfect information, stronger privacy induces the seller to broaden his product design. The rotation in the pricing schedule described above implies that, for small Δ , profitability hinges on how design changes affect the most misaligned consumers within each posterior. Since low privacy implies the seller sells to the entire posterior support, those least well-matched impose the binding constraint on design. As Δ rises and the posterior widens, this constraint tightens: it becomes harder to profitably serve consumers with strong preference mismatches. To retain them, the seller broadens product design to accommodate a wider range of tastes and maintain demand across the full posterior. This is consistent with the behavior illustrated in [Figure 5](#): as privacy increases slightly from $\Delta = 0$, the optimal product design s^* decreases.

At higher privacy levels, however, this relationship can reverse: the seller may find it optimal to increase product design. In this region, price and product design become strategic complements:

holding design constant no longer produces a uniform shift in the price schedule, and changing the design feeds back into optimal pricing in a non-trivial way. [Proposition 5](#) formalizes this result by characterizing the sufficient condition for such a reversal.

Proposition 5 (Intermediate- Δ comparative statics) *Suppose $\Delta < \theta_{r,\text{feas}}(s)/3$, so that the acceptance-inside regime is empty and only sell-to-all and partial-overlap regions are relevant. Then*

$$\frac{\partial^2 \Pi(s(\Delta), \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \left[-c'_v(1-s) + \frac{7}{3}t\Delta \right],$$

so that

$$\frac{\partial^2 \Pi(s(\Delta), \Delta)}{\partial s \partial \Delta} > 0 \iff \Delta > \frac{3}{7t} c'_v(1-s).$$

Feasibility of this region further requires

$$\frac{3}{7t} c'_v(1-s^*) < \Delta < \frac{V - mc - c_v(1-s^*)}{2ts^*} = \frac{\theta_{r,\text{feas}}}{2}.$$

[Proposition 5](#) shows that with intermediate privacy levels, the cross-partial derivative decomposes into a vertical component, $-c'_v(1-s)$, and a horizontal component, $\frac{7}{3}t\Delta$. The vertical term is negative and independent of Δ , reflecting that although increasing s always reduces consumers' vertical mismatch, the strength of this benefit does not change with privacy. In contrast, the horizontal term is positive and scales linearly with Δ , capturing that as uncertainty about the consumer's preferences increases, the marginal impact of s on expected horizontal match quality and profit becomes stronger. When the negative vertical term dominates, an increase in privacy weakens the profitability of raising s . When the positive horizontal effect dominates (e.g. when privacy is sufficiently strong), an increase in privacy strengthens the profitability of a more specialized product design, reversing the relationship established in [Proposition 4](#).

The shift in the sign of the cross-partial reflects a change in the composition of served customers. At an intermediate level of privacy, the posterior $\mathcal{S}(x, \Delta)$ widens and the seller no longer finds it profitable to serve all consumers within each posterior. Instead, the seller focuses on consumers whose tastes are relatively close to his product and excludes those with greater mismatch. Because the served consumers are more homogeneous and exhibit lower horizontal dispersion, the price response to design becomes directionally uniform: increasing (decreasing) design raises (lowers) prices for the buyers. In this regime, price and design act as strategic complements. The seller can profitably choose a more niche (higher- s) design and raise prices without losing much demand.

Taken together, these results show that the seller's optimal product design is non-monotonic in privacy: it decreases near $\Delta = 0$ and increases for sufficiently large Δ . The following theorem

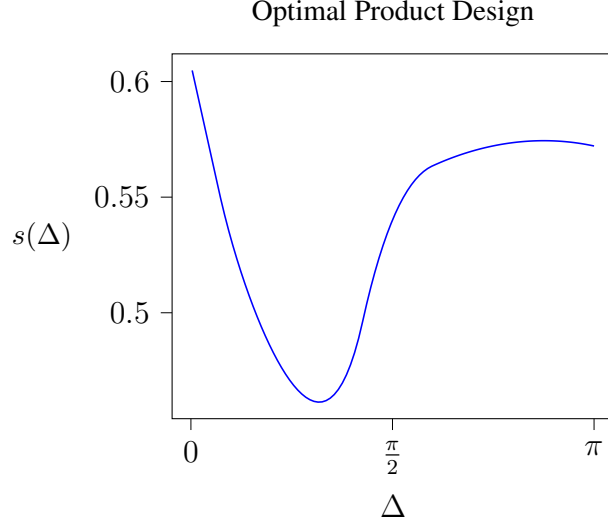


Figure 5: Optimal product design s^* as a function of privacy Δ . The non-monotonic relationship reflects the shift in the monopolist's strategy from broadening to niche targeting. Parameters: $V = 1.25$, $mc = 0$, $t = 0.5$, $k = 1$, and cost function $c_v(1 - s) = k(1 - s)^2$.

formalizes this comparative-statics result.

Theorem 1 (Non-monotonicity of optimal design) Let $s^*(\Delta)$ denote the profit-maximizing design at privacy level Δ . Suppose $s^*(\Delta)$ is interior and satisfies the second-order condition $\frac{\partial^2 \Pi}{\partial s^2}|_{s^*} < 0$. Assume $\theta_{r, \text{feas}}(s^*(0)) > 0$. Then:

1. (Near $\Delta = 0$) By [Proposition 4](#), $\frac{ds^*}{d\Delta}(0^+) < 0$.
2. (Intermediate Δ) If the conditions of [Proposition 5](#) are satisfied for some $\Delta^+ \in (0, \pi]$, then $\frac{ds^*}{d\Delta}(\Delta^+) > 0$.

Consequently, whenever this intermediate- Δ range exists, $s^*(\Delta)$ is non-monotonic in Δ : it decreases near $\Delta = 0$ and increases for sufficiently large Δ .

[Theorem 1](#) establishes that $s^*(\Delta)$ is non-monotonic whenever the intermediate Δ region of [Proposition 5](#) is nonempty.³ Under the quadratic vertical mismatch cost $c_v(1 - s) = (1 - s)^2$, the following corollary provides an explicit sufficient condition for this region.⁴

Corollary 1 (Sufficient condition under quadratic vertical mismatch cost) If $c_v(1 - s) = (1 -$

³I verify numerically that, for the parameter region considered, $s^*(\Delta)$ is an interior maximizer satisfying $\frac{\partial^2 \Pi}{\partial s^2}|_{s^*} < 0$ throughout; see [Appendix A.3](#).

⁴[Appendix A.6](#) provides a numerical illustration of the effect of increasing privacy with weak and intermediate levels of privacy.

$s)^2$, then $c'_v(1 - s) = 2(1 - s)$. For any $s \in [s_{\min}, 1]$, the cross-partial is positive whenever

$$\Delta > \frac{6}{7t}(1 - s_{\min}),$$

and the intermediate- Δ region (sell-to-all and partial-overlap only) additionally requires

$$\Delta < \bar{\Delta} \equiv \inf_{s \in [s_{\min}, 1]} \frac{\theta_{r, \text{feas}}(s)}{3} = \min \left\{ \frac{V - mc}{3t}, \frac{V - mc - (1 - s_{\min})^2}{3ts_{\min}} \right\}.$$

This interval is nonempty provided

$$\frac{6}{7t}(1 - s_{\min}) < \bar{\Delta},$$

which holds when $s_{\min}$ is bounded away from 0 or $V - mc$ is sufficiently large; in that case any Δ in the interval yields $\frac{\partial^2 \Pi}{\partial s \partial \Delta} > 0$ and hence $\frac{ds^*}{d\Delta} > 0$.

While [Theorem 1](#) and [Corollary 1](#) characterize the non-monotonicity at weak and intermediate privacy levels, this upward trend does not continue indefinitely for stronger levels of privacy.

Strong Privacy Regulation. When privacy becomes even stronger, the firm's ability to target consumers deteriorates further, and the pattern observed at intermediate levels gradually reverses. The posterior $\mathcal{S}(x, \Delta)$ spans nearly the entire circle, and the seller can no longer target a distinct group of consumers with similar tastes. Most signals now correspond to the partial-overlap and acceptance-inside-posterior regimes. In the latter region, the firm continues to serve the least aligned consumers it can still reach, who exert a mild downward pressure on design, since further specialization increases mismatch without generating meaningful price gains. As a result, the strong complementarity between price and design that characterized intermediate privacy levels gradually weakens. The optimal design flattens out and may decline marginally, reflecting that when preferences are highly dispersed under strong privacy, the seller faces a nearly uniform market and has little scope for targeted specialization. This decline is visible for Δ near π in [Figure 5](#).

5.2 Consumer Surplus.

So far, the analysis has focused on the seller's price and product design decisions. However, privacy regulation also directly affects consumers. To assess its welfare implications, I analyze consumer surplus within each geometric regime and then aggregate across signals, paralleling the cross-partial decomposition in [Section 5.1.1](#). This highlights how the seller's incentives align with (or diverge from) consumer welfare, and identifies which privacy levels are most beneficial for consumers.

As in the cross-partial decomposition, consumer surplus takes a different closed-form solution in each of the three regimes (sell-to-all, acceptance-inside-posterior, and partial-overlap), derived in [Appendix A.4](#). Aggregating the signal-level surplus over x and splitting across the three regimes gives consumer surplus under an arbitrary design,

$$CS(s, \Delta) = \frac{1}{\pi} \int_0^\pi CS(s, x, \Delta) dx = \sum_{R \in \{\text{all}, \text{int}, \text{par}\}} \frac{1}{\pi} \int_{\mathcal{X}_R(s, \Delta)} CS_R(s, x, \Delta) dx.$$

Evaluating at the seller's optimal design defines the reduced-form consumer surplus at each privacy level,

$$CS(\Delta) = CS(s^*(\Delta), \Delta),$$

where the integral is taken over $x \in [0, \pi]$ by the symmetry of the circle, so that $d_0(x, 0) = x$ throughout.

The next proposition characterizes how privacy influences per-signal consumer surplus within this aggregation, showing how its response to a change in privacy varies across regimes.

Proposition 6 (Per-signal $\partial_\Delta CS(s, x, \Delta)$) *For each signal $x \in [0, \pi]$, the per-regime consumer surplus $CS_R(s, x, \Delta)$, where $R \in \{\text{all}, \text{int}, \text{par}\}$ has partial derivatives with respect to Δ :*

$$\begin{aligned} \partial_\Delta CS_{\text{all}} &= ts > 0, \\ \partial_\Delta CS_{\text{int}} &= -\frac{(V - c_v(1 - s) - mc)^2}{8ts\Delta^2} < 0, \\ \partial_\Delta CS_{\text{par}} &= \frac{ts}{16\Delta^2} \left(\frac{V - c_v(1 - s) - mc}{ts} + \Delta - x \right) \left(\Delta - \frac{V - c_v(1 - s) - mc}{ts} + x \right) > 0. \end{aligned}$$

At the per-signal level, consumer surplus exhibits opposing behaviors across regimes: it is strictly increasing in privacy within the sell-to-all and partial-overlap regimes, but strictly decreasing in the acceptance-inside-posterior regime. Thus, the aggregate shape of $CS(s, \Delta)$ depends on how probability mass over signals is allocated across these regimes as Δ varies. The next proposition characterizes the initial slope of consumer surplus in the absence of privacy regulation, providing the baseline for how privacy begins to affect consumers.

Proposition 7 (Consumer Surplus near $\Delta = 0$) *Let $s^*(\Delta)$ denote an optimal design and assume $\theta_{r, \text{int}}(s) > 0$. Then*

$$\left. \frac{d}{d\Delta} CS(s^*(\Delta), \Delta) \right|_{\Delta=0^+} = ts^*(0) > 0.$$

When Δ is sufficiently small, [Proposition 7](#) shows that consumer surplus is strictly increasing in privacy. Intuitively, under perfect information consumers obtain no surplus; a small amount of

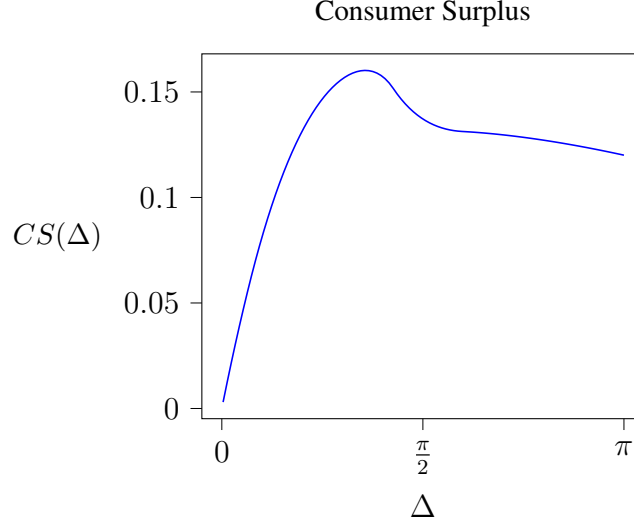


Figure 6: Consumer surplus as a function of privacy Δ . The non-monotonic relationship, initially increasing then decreasing, reflects the changing balance in the monopolist's strategic incentives between price competition and product design. Evaluated at the optimal product design $s^*(\Delta)$ from Figure 5. Parameters: $V = 1.25$, $mc = 0$, $t = 0.5$, $k = 1$, and cost function $c_v(1 - s) = k(1 - s)^2$.

noise softens prices and creates informational rents. This is consistent with the upward slope in Figure 6 near the origin.

As privacy continues to rise, however, the relationship between privacy and consumer surplus reverses. The same mechanism that generates the non-monotonicity in the seller's optimal product design also governs the shape of consumer welfare. However, consumer surplus depends on how consumers are affected by both the prices they face, and whether they are served at all. Weak privacy regulation limits the seller's ability to perfectly price discriminate. To retain marginal consumers, the seller broadens his product design and reduces prices for those with higher horizontal mismatch, increasing aggregate consumer surplus at the cost of a small contraction in market coverage.

Once privacy reaches an intermediate level, however, product design and price become strategic complements. The seller shifts his focus toward consumers whose preferences are closely aligned with his product, offering a more niche design and charging higher prices. As Δ increases further, the posterior widens in both directions: the firm continues to serve well-matched consumers while excluding those with greater preference mismatch. Both the exclusion of these consumers and the higher prices faced by those still served reduce aggregate consumer surplus.

Consequently, consumer surplus can be non-monotonic: initially increasing, but eventually decreasing in privacy, as shown in Figure 6. In this case, consumers benefit most from an intermediate level of privacy. Too little privacy allows the seller to extract nearly all rents, whereas too much privacy induces the seller to adopt a niche design that serves a narrow segment at higher prices

while excluding others. While the pattern in [Figure 6](#) illustrates a non-monotonic relationship between consumer surplus and privacy, this outcome should be interpreted as a possibility rather than a general result. The shape of $CS(\Delta)$ depends on the seller's product design $s(\Delta)$, which in turn is sensitive to the underlying parameter values that govern how prices and market coverage respond to changes in privacy.

This consumer welfare result suggests that stronger privacy regulation is not always beneficial for consumers. When the seller's product design responds non-monotonically to privacy, pushing regulation beyond an intermediate level can reduce consumer welfare – both by inducing a more niche product that excludes poorly-matched consumers and by weakening the informational friction that had previously constrained the firm's ability to extract surplus. This highlights the importance of accounting for firm-side responses to privacy regulation. The indirect effects on product design can shape consumer welfare in ways that are not immediately apparent from a consumer-centric analysis alone.

5.3 Welfare

The seller's profit-maximizing product design does not necessarily maximize total welfare. In particular, there are two distinct distortions that separate it from the efficient outcome. First, the seller chooses a product design to maximize profit rather than valuing match quality directly – a design distortion. Second, personalized pricing excludes consumers whose match value exceeds marginal cost, but whose willingness to pay falls below the personalized price – a pricing distortion. To separate these, I consider two welfare benchmarks. The first-best outcome removes both distortions: the social planner chooses a design that maximizes gross match value net of cost, with no transfers. The second-best outcome isolates the pricing distortion: the planner chooses the product design, but the seller retains personalized pricing. Comparing the two benchmarks to the profit-maximizing equilibrium decomposes the total welfare loss into a component due to design and a component due to pricing.

5.3.1 First-Best Outcome

With no price transfers, the social planner chooses the optimal product design that maximizes the gross match value net of costs. The per-buyer gross surplus is therefore

$$V - c_v(1 - s) - c_h(s) - mc.$$

Let $\theta_{FB}(s, x, \Delta) = \min\{\theta_{r, \text{feas}}, x + \Delta\}$ denote the planner's acceptance reach, such that $\mathcal{R}(\theta_{FB}) = \{x : d_0(x, 0) \leq \theta_{FB}\}$. Thus, the set of buyers is then

$$B_{FB}(x, \Delta, r) = S(x, \Delta) \cap \mathcal{R}(\theta_{FB}(s, x, \Delta)).$$

Hence, the per-signal welfare is

$$w_{FB}(s, x, \Delta) = \frac{1}{2\Delta} \int_{B_{FB}(x, \Delta, r)} (V - c_v(1 - s) - ts\theta - mc) d\theta.$$

The first-best objective is

$$W_{FB}(s, \Delta) = \frac{1}{\pi} \int_0^\pi w_{FB}(s, x, \Delta) dx, \quad s^{FB}(\Delta) \in \operatorname{argmax}_{s \in [s_{\min}, 1]} W_{FB}(s, \Delta). \quad (13)$$

5.3.2 Second-Best Outcome

If the social planner cannot control prices but can still choose the product design, the outcome is second-best. The seller continues to set personalized prices based on the realized signal x , while the planner selects s to maximize total welfare given this pricing behavior.

Total welfare is the sum of consumer surplus and the seller's profit:

$$W(\Delta) = CS(\Delta) + \Pi(\Delta). \quad (14)$$

The second-best product design solves

$$s^{SB}(\Delta) \in \operatorname{argmax}_{s \in [s_{\min}, 1]} W(s, \Delta) = \operatorname{argmax}_{s \in [s_{\min}, 1]} [\Pi(s, \Delta) + CS(s, \Delta)]. \quad (15)$$

The planner thus internalizes the welfare effects of product design on both profits and consumer surplus, but cannot eliminate the pricing distortion created by the seller's personalized prices.

5.3.3 Welfare Decomposition

The gap between the first- and second-best outcome therefore measures the pricing distortion, while the gap between the second-best and the profit-maximizing outcome measures the design distortion. In particular,

$$\underbrace{W_{FB}(s^{FB}, \Delta) - W(s^*, \Delta)}_{\text{total welfare loss}} = \underbrace{[W_{FB}(s^{FB}, \Delta) - W(s^{SB}, \Delta)]}_{\text{pricing distortion}} + \underbrace{[W(s^{SB}, \Delta) - W(s^*, \Delta)]}_{\text{design distortion}}. \quad (16)$$

Each term is nonnegative: the first-best optimizes over design with no pricing distortion, the second-best over the same design choice subject to the seller's prices, and the equilibrium fixes design at the profit-maximizing level. This yields the welfare ordering $W_{FB}(s^{FB}, \Delta) \geq W(s^{SB}, \Delta) \geq W(s^*, \Delta)$, with the first inequality strict whenever some consumers served under the first-best are excluded under personalized pricing.

5.3.4 Welfare Comparative Statics

Since the first-best and second-best objectives aggregate surplus across the same regime-dependent structure as the profit function, closed-form solutions for s^{FB} and s^{SB} are generally not available. The following discussion is therefore based on numerical evaluation, illustrated in [Figure 7](#).

Product Design Choice. As shown in the top panel of [Figure 7](#), the first-best product design is always more niche than the seller's profit maximizing design because the planner values consumers' match quality directly, whereas the seller only values it through its effect on demand and prices. The seller broadens design to expand reach and extract higher profits, while the planner narrows design to improve total match efficiency, even at the cost of excluding marginal consumers.

In the second-best, the planner now faces the same price schedule as the seller. Consequently, both $s^*(\Delta)$ and $s^{SB}(\Delta)$ exhibit similar non-monotonic responses to increases in privacy, reflecting the underlying price-design mechanism discussed in [Section 5.1.2](#). Since the social planner values consumer surplus in addition to profit, $s^{SB}(\Delta)$ generally is more niche than $s^*(\Delta)$. However, near the privacy level where the relationship between price and product design changes, the planner's product design begins to increase at a higher Δ than the seller's, generating a narrow range of privacy levels where $s^{SB}(\Delta) < s^*(\Delta)$. Intuitively, the planner internalizes the welfare cost of higher prices: when price and design become complements, the seller immediately raises both to exploit the higher markups, while the planner adjusts more cautiously, recognizing that the same complementarity amplifies the price distortion that reduces consumer surplus.

Total Welfare. Consistent with the decomposition in [\(16\)](#), the second-best welfare curve lies between the first-best and profit-maximizing curves and closely tracks the latter, highlighting that the main source of inefficiency is the pricing distortion created by personalized markups, not the choice of product design. Because the planner and the seller face the same price schedule in the second-best, adjusting design alone cannot fully offset the surplus loss created by personalized markups. Still, shifting from $s^*(\Delta)$ to $s^{SB}(\Delta)$ raises total welfare. The remaining gap between the first-best and second-best welfare therefore quantifies the welfare cost of the seller's personalized pricing.

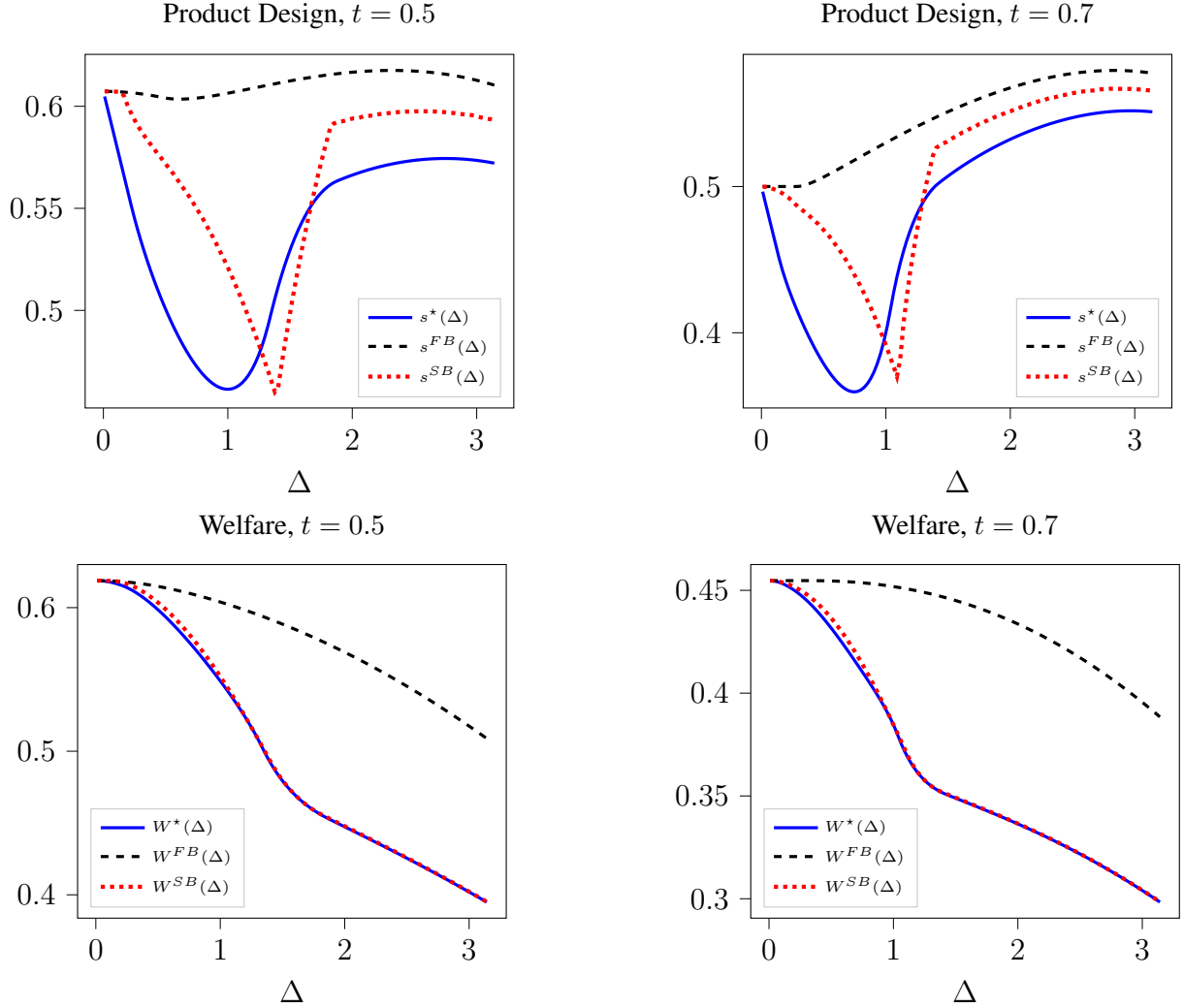


Figure 7: Comparison of first-best (dashed black), second-best (dotted red), and profit-maximizing (solid blue) product design and welfare as a function of privacy Δ . The first-best product design is the most niche, while the second-best follows a similar trend to the profit-maximizing design. First-best welfare is highest by a substantial margin, while second-best welfare closely tracks welfare under profit maximization. Parameters: $V = 1.25$, $mc = 0$, $t = 0.5$, $k = 1$, and cost function $c_v(1 - s) = k(1 - s)^2$.

Horizontal Mismatch Cost. Finally, comparing product design and welfare across values of the horizontal mismatch parameter shows that as t increases, product design overall becomes broader. This pattern is intuitive: when traveling along the circle becomes more costly for consumers, niche designs provide less additional surplus because fewer consumers are willing to bear the higher mismatch disutility. Both the seller and planner thus choose broader product design to appeal to a wider set of consumers.

While both the seller and planner respond to higher t with broader designs, the parameter t also uniquely shapes how the first-best product design s^{FB} responds to privacy. When horizontal mismatch costs are small, the planner puts greater weight on minimizing vertical costs. As privacy increases and the posterior widens, the vertical concerns dominate and the first-best product design broadens. As t rises, the planner becomes more willing to tolerate a higher vertical cost if doing so reduces horizontal losses. Consequently, for intermediate t , the first-best product design can be non-monotonic: the planner broadens product design at first but eventually switches to more niche products as horizontal losses become severe at large Δ . For very high t , horizontal mismatch dominates entirely, and s^{FB} increases in Δ as the planner focuses on maintaining high match quality among the remaining well-matched consumers.

6 Conclusion

This paper studies the impact of privacy regulation on a seller's strategic decisions over product design and pricing. The analysis focuses on a setting where the seller has sufficient market power to commit to product design without competitive pressure. Using a modified [Salop \(1979\)](#) framework, I find that an intermediate level of privacy is optimal for consumers. While this result mirrors a common finding in the privacy literature, the underlying mechanism in this model is distinct: it operates through the firm's ex-ante product design choice. Specifically, the seller's product design can respond non-monotonically to privacy. At intermediate privacy levels, this leads the firm to strategically choose a more niche design, which increases consumers' average match utility enough to outweigh the accompanying increase in price.

As privacy increases, the firm initially broadens product design to retain far-away consumers as information becomes noisier. At intermediate privacy levels, as the firm serves a more local market, design and price become strategic complements, leading to a more niche product choice. When privacy becomes extreme and the firm can no longer target consumers effectively, product design broadens to appeal to the entire market. This pattern contrasts with the common presumption that greater privacy uniformly pushes firms toward broader, less targeted products. The non-monotonicity arises because the firm must commit to product design before observing consumer signals, so privacy shifts which consumers determine the design incentives.

Beyond the seller’s behavior, I also compare market outcomes to the socially optimal benchmarks. The first-best product design is always more niche than the seller’s, as the planner internalizes consumers’ match quality rather than valuing it only through demand and pricing. In the second-best, when price distortions remain present, both the planner and the seller can exhibit similar non-monotonic responses to privacy, though the planner chooses slightly more niche products for most privacy levels. These results highlight that welfare differences arise primarily from how information affects price discrimination, rather than from misaligned incentives over product design. This suggests that privacy regulation alone may be insufficient. Effective policy should also address firms’ use of detailed consumer information to implement personalized pricing strategies.

I plan to extend this framework to a setting with an endogenous number of firms, relaxing the monopolistic structure of the current model. Allowing for competition and entry decisions would link privacy regulation not only to product design and pricing, but also to market structure. This extension would be related to the literature on specialization and competition (e.g., [González-Maestre and López-Cerdá \(2018\)](#)) and yield richer policy implications, particularly around how privacy regulation may affect entry, competition, and consumer welfare in differentiated markets. This extension would facilitate closer comparison to empirical work showing that privacy regulation can reshape firm behavior and market outcomes, such as [Johnson et al. \(2023\)](#) on market concentration. Another natural extension is to incorporate dynamics. In multi-period environments, firms may accumulate information over time, and privacy regulation can constrain how past signals are stored or used in future periods. Such dynamics may generate intertemporal incentives in pricing and product design that are absent in a static setting, providing additional insight into how privacy affects long-run firm behavior and consumer welfare.

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A Appendix

A.1 Closed forms for the measure of buyers

$$L(x, \Delta, r) = \begin{cases} 0, & x \geq \theta_r + \Delta, \\ 2\theta_r, & \theta_r \leq \Delta \text{ and } x \leq \Delta - \theta_r, \\ 2\Delta, & \theta_r \geq \Delta \text{ and } x \leq \theta_r - \Delta, \\ \theta_r + \Delta - d_0(x, 0), & |\theta_r - \Delta| < x < \theta_r + \Delta. \end{cases}$$

The buyer measure is always bounded by

$$L(x, \Delta, \theta_r) = \max\{0, \min\{\theta_r, x + \Delta, \pi\} - \max\{0, x - \Delta\}\}.$$

A.2 Closed forms for the profit cross-partial

To characterize how privacy affects optimal design, I compute the cross-partial under the three feasible optimal-acceptance regimes identified in [Section 4.1](#). The following lemmas derive the cross-partial for each regime.

Lemma 4 (Cross-partial: Posterior inside Acceptance ($S \subset \mathcal{R}$)) *If $\theta_r^*(s, x, \Delta) = \theta_{r,\text{all}}^*(x, \Delta) = x + \Delta$ is feasible, then the posterior lies within acceptance. Let $\mathcal{X}_{\text{all}}(s, \Delta) \subseteq [0, \pi]$ denote the set of x where this regime is optimal.*

Then,

$$\frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{all}}(s, \Delta)} (-t) dx.$$

Lemma 5 (Cross-partial: Acceptance inside Posterior ($\mathcal{R} \subset S$)) *When $\theta_r^*(s, x, \Delta) = \theta_{r,\text{int}}^* \leq \Delta - x$ (with $0 \leq x \leq \Delta$) is feasible, the acceptance lies fully within the posterior. Let $\mathcal{X}_{\text{int}}(s, \Delta) \subseteq [0, \pi]$ be the set of x where this regime is optimal. Then*

$$\frac{\partial^2 \Pi_{\text{int}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{int}}(s, \Delta)} -\frac{\theta_{r,\text{int}}^*}{\Delta^2} \left(c'_v(1-s) - t \theta_{r,\text{int}}^* \right) dx,$$

Lemma 6 (Cross-partial: Partial overlap ($S \cap \mathcal{R} \neq \emptyset$)) *If feasible, consider signals x for which the acceptance region and posterior partially overlap, so $\theta_r^*(s, x, \Delta) = \theta_{r,\text{par}}^*$. Let $\mathcal{X}_{\text{par}}(s, \Delta) \subseteq$*

$[0, \pi]$ denote the set of x where this regime is optimal. Then,

$$\frac{\partial^2 \Pi_{\text{par}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{par}}(s, \Delta)} \left[\frac{(c'_v(1-s) - t\theta_{r,\text{par}}^*)}{4\Delta^2} \left(x - \frac{V - c_v(1-s) - mc}{ts} \right) + \frac{L(x, \Delta, \theta_{r,\text{par}}^*)t}{4\Delta} \right] dx.$$

In the sell-to-all regime (Lemma 4), the cross-partial is unambiguously negative: when the firm serves the entire posterior, greater privacy strictly weakens the incentive to specialize. In the remaining two regimes, the effect of privacy on the incentive to specialize is ambiguous. Collecting the expressions from Lemma 4–Lemma 6 gives the total cross-partial effect of (s, Δ) . Since the product design s is chosen ex-ante and does not vary with the signal x , the overall cross-partial of profit is obtained by aggregating the per-signal cross-partials across all x . This decomposition is formalized in Proposition 3.

A.3 Numerical Verification of the Second-Order Condition

Theorem 1 takes as a hypothesis that $s^*(\Delta)$ is an interior maximizer satisfying the second-order condition $\frac{\partial^2 \Pi}{\partial s^2}|_{s^*(\Delta)} < 0$. Because the per-signal profit $\Pi^*(s, x, \Delta)$ is piecewise across the three trade regimes and each piece is convex in s , the objective $\Pi(s, \Delta)$ is not globally concave, and the second-order condition cannot be established by a curvature argument on the pieces. I therefore verify it numerically at the calibrated parameters.

For each Δ , I evaluate $\Pi(s, \Delta)$ on a fine grid over $s \in [s_{\min}, 1]$, recomputing the per-signal optimum directly from primitives rather than from the closed-form regime optima, so the check is independent of Lemma 3. As Figure 8a shows, the objective is single-peaked for every Δ , with a global maximizer strictly interior to $[s_{\min}, 1]$. Figure 8b plots $\frac{\partial^2 \Pi}{\partial s^2}$ at $s^*(\Delta)$; it is negative throughout, with a maximum of -0.61 over the grid, so the second-order condition holds with margin at every privacy level, including the trough region of Figure 5. As a further check, this brute-force computation matches the analytic integrator used for the paper’s figures to within 5×10^{-4} .

These checks confirm the hypothesis of Theorem 1 for the parameter region considered. They do not establish the second-order condition globally: at other configurations (for example, higher V/mc or a linear vertical cost) the optimum can lie at the boundary $s = s_{\min}$, which the theorem excludes.

A.4 Closed forms for the consumer surplus cross-partial

Lemma 7 (Consumer Surplus: Sell-to-all ($S \subseteq \mathcal{R}$)) *If $\theta_r^* = \theta_{r,\text{all}}^*$ is feasible, then*

$$CS_{\text{all}}(s, x, \Delta) = ts\Delta.$$

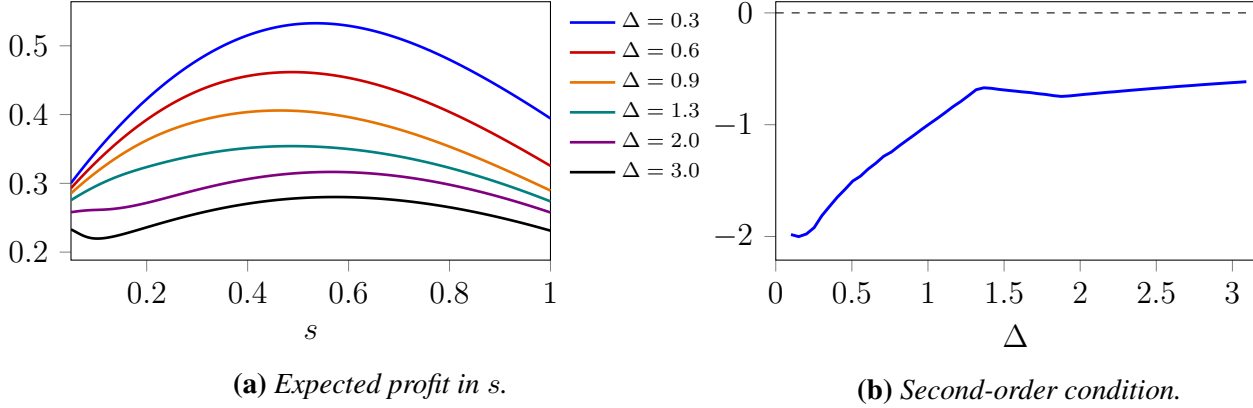


Figure 8: Numerical verification of the interior optimum and second-order condition. Parameters: $V = 1.25$, $mc = 0$, $t = 0.5$, $k = 1$.

Lemma 8 (Consumer Surplus: Acceptance inside posterior ($\mathcal{R} \subseteq S$)) If $\theta_r^* = \theta_{r,\text{int}}^*(s)$ is feasible, then

$$CS_{\text{int}}(s, x, \Delta) = \frac{(V - c_v(1 - s) - mc)^2}{8ts\Delta}.$$

Lemma 9 (Consumer Surplus: Partial overlap ($S \cap \mathcal{R} \neq \emptyset$)) If $\theta_r^* = \theta_{r,\text{par}}^*(s, x, \Delta)$ is feasible, then

$$CS_{\text{par}}(s, x, \Delta) = \frac{ts}{4\Delta} \left(\frac{V - c_v(1 - s) - mc}{2ts} + \frac{\Delta - x}{2} \right)^2.$$

A.5 Omitted Proofs

A.5.1 Proof of Proposition 1

Using (2), (3), (5), and (6), we compare the benchmark-optimal product designs under no information and perfect information. Throughout, assume c_v is increasing and convex. Let s_{full}^N and s_{part}^N denote the optimal product design under full and partial market coverage, respectively, when there is no information. Similarly, s_{full}^P and s_{part}^P denote the optimal product design under full and partial market coverage, respectively, when there is full information.

Full market coverage. The full-coverage first-order conditions are

$$c'_v(1 - s_{\text{full}}^N) = t\pi \quad \text{and} \quad c'_v(1 - s_{\text{full}}^P) = \frac{t\pi}{2}.$$

Since $t\pi > \frac{t\pi}{2}$ and c'_v is increasing, it follows that

$$1 - s_{\text{full}}^N > 1 - s_{\text{full}}^P,$$

and hence

$$s_{\text{full}}^P > s_{\text{full}}^N.$$

Partial market coverage. In the partial-coverage regime, substituting the equilibrium demand expressions $Q_N(s) = \frac{V - c_v(1-s) - mc}{2ts\pi}$ into (3) and substituting $Q_P(s) = \frac{V - c_v(1-s) - mc}{ts\pi}$ into (6) gives the same condition

$$c'_v(1-s) = \frac{V - c_v(1-s) - mc}{2s}.$$

Therefore, the partial-coverage candidate designs coincide:

$$s_{\text{part}}^P = s_{\text{part}}^N.$$

A.5.2 Proof of Lemma 3

Throughout, write $\bar{V} \equiv V - c_v(1-s)$ and recall the price rule $p(\theta_r) = \bar{V} - ts\theta_r$ and the maximum feasible reach $\theta_{r,\text{feas}} = \frac{\bar{V} - mc}{ts}$. For $x \in [0, \pi]$ and $\Delta > 0$, the partial-overlap interval has lower endpoint $\max\{0, \Delta - x, x - \Delta\} = \max\{0, |x - \Delta|\} = |x - \Delta|$, so the three regimes partition $[0, \theta_{r,\text{feas}}]$ as

$$\underbrace{[0, |x - \Delta|]}_{\text{interior, if } x \leq \Delta} \cup \underbrace{[|x - \Delta|, x + \Delta]}_{\text{partial}} \cup \underbrace{[x + \Delta, \theta_{r,\text{feas}}]}_{\text{all}},$$

each intersected with $[0, \theta_{r,\text{feas}}]$. I treat each regime separately, then verify continuity at the shared boundaries.

(i) *Sell-to-all* ($\theta_r \geq x + \Delta$). The acceptance region covers the posterior, so $L = 2\Delta$ and the buyer share is $Q = L/2\Delta = 1$. Profit is $\Pi(\theta_r) = \bar{V} - mc - ts\theta_r$, which is linear and strictly decreasing in θ_r (slope $-ts < 0$). The maximum over this segment is attained at its left endpoint $\theta_{r,\text{all}}^* = x + \Delta$, the smallest reach that still covers $S(x, \Delta)$. Substituting gives $p_{\text{all}} = \bar{V} - ts(x + \Delta)$ and $\Pi_{\text{all}}^* = \bar{V} - mc - ts(x + \Delta)$. This candidate is feasible iff $p_{\text{all}} \geq mc$, i.e. $x + \Delta \leq \theta_{r,\text{feas}}$.

(ii) *Interior* ($\theta_r < \Delta$, requires $x \leq \Delta$). Here $L = 2\theta_r$, so $\Pi(\theta_r) = \frac{(\bar{V} - mc - ts\theta_r)\theta_r}{\Delta}$. This is a concave quadratic in θ_r , since $\Pi''(\theta_r) = -\frac{2ts}{\Delta} < 0$. The first-order condition $\Pi'(\theta_r) = \frac{\bar{V} - mc - 2ts\theta_r}{\Delta} = 0$ gives the unique stationary point

$$\theta_{r,\text{int}}^* = \frac{\bar{V} - mc}{2ts},$$

which is the global maximizer by concavity. The price rule yields $p_{\text{int}}^* = \bar{V} - ts\theta_{r,\text{int}}^* = \frac{1}{2}(\bar{V} + mc)$,

and substituting back gives $\Pi_{\text{int}}^* = \frac{(\bar{V} - mc)^2}{4ts\Delta}$. The interior point lies in the regime iff $0 \leq \theta_{r,\text{int}}^* \leq \Delta - x$; the lower bound is automatic since $\bar{V} - mc \geq 0$, and the upper bound is equivalent to $x \leq \Delta - \frac{\bar{V} - mc}{2ts}$. If this fails (the signal is too far from the firm), Π is increasing up to the right endpoint $\theta_r = \Delta - x$ (when $x < \Delta$), so the best outcome on this segment is the boundary

$$p_{\text{in,bd}} = \bar{V} - ts(\Delta - x), \quad \Pi_{\text{in,bd}} = \frac{(\bar{V} - mc - ts(\Delta - x))(\Delta - x)}{\Delta},$$

at which the geometry transitions into the partial-overlap regime.

(iii) *Partial overlap* ($|x - \Delta| < \theta_r < x + \Delta$). Here $L = \theta_r + \Delta - x$, so $\Pi(\theta_r) = \frac{(\bar{V} - mc - ts\theta_r)(\theta_r + \Delta - x)}{2\Delta}$.

This is again concave, with $\Pi''(\theta_r) = -\frac{ts}{\Delta} < 0$. The first-order condition gives the unique stationary point

$$\theta_{r,\text{par}}^* = \frac{\bar{V} - mc - ts(\Delta - x)}{2ts} = \frac{\bar{V} - mc}{2ts} - \frac{\Delta - x}{2},$$

i.e. the interior maximizer of regime (ii) shifted by $-\frac{1}{2}(\Delta - x)$. The price rule gives $p_{\text{par}}^* = \bar{V} - ts\theta_{r,\text{par}}^*$, and substitution yields

$$\Pi_{\text{par}}^* = \frac{(\bar{V} - mc + ts(\Delta - x))^2}{8ts\Delta}.$$

This candidate is feasible iff $|x - \Delta| < \theta_{r,\text{par}}^* < x + \Delta$ and $\theta_{r,\text{par}}^* \leq \theta_{r,\text{feas}}$. Writing the first pair of inequalities in terms of $\frac{\bar{V} - mc}{ts}$ gives

$$\begin{cases} x \leq \Delta : & 3(\Delta - x) < \frac{\bar{V} - mc}{ts} < x + 3\Delta, \\ x > \Delta : & x - \Delta < \frac{\bar{V} - mc}{ts} < x + 3\Delta. \end{cases}$$

Continuity at the boundaries. At $\theta_r = \Delta - x$ the interior and partial profit expressions coincide: both equal $\frac{(\bar{V} - mc - ts(\Delta - x))(\Delta - x)}{\Delta}$ (the partial length $\theta_r + \Delta - x$ equals $2\theta_r = 2(\Delta - x)$ there). At $\theta_r = x + \Delta$ the partial and sell-to-all expressions coincide: the partial length $\theta_r + \Delta - x$ equals 2Δ , so both reduce to $\bar{V} - mc - ts(x + \Delta)$. Hence $\Pi(\cdot \mid s, x, \Delta)$ is continuous across the two regime boundaries.

A.5.3 Proof of Proposition 2

By Lemma 3, the three regimes partition the feasible interval $[0, \theta_{r,\text{feas}}]$, and on each piece $\Pi(\theta_r \mid s, x, \Delta)$ is concave (parts (i)–(iii) give slopes/second derivatives that are weakly negative). By

the continuity established above, the piecewise objective is continuous on the whole interval. A continuous function that is concave on each piece of a finite partition attains its maximum either at an interior stationary point of some piece or at an endpoint of the feasible interval. The interior stationary points are exactly $\theta_{r,\text{all}}^*$, $\theta_{r,\text{int}}^*$, and $\theta_{r,\text{par}}^*$ (each included only when it lies in its regime), and the only binding endpoint is $\theta_{r,\text{feas}}$, since $\theta_r = 0$ yields zero profit and the internal boundaries are covered by continuity. Comparing Π across the candidate set Θ therefore returns the global maximizer $\theta_r^*(s, x, \Delta)$, with $p^* = p(\theta_r^*)$.

A.5.4 Proof of Lemma 4.

If $\theta_r^*(s, x, \Delta) = \theta_{r,\text{all}}(x, \Delta) = x + \Delta$, then the posterior lies within acceptance. In this case $L(x, \Delta, \theta_r) = 2\Delta$ and $\Pi(r) = V - c_v(1 - s) - mc - ts\theta_r$. Lemma 3 gives $\theta_r^*(s, x, \Delta) = x + \Delta$ on the set $\mathcal{X}_{\text{all}}(s, \Delta) \subseteq [0, \pi]$ where this regime is optimal. Hence the contribution of this regime to the cross-partial is

$$\frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{all}}(s, \Delta)} \partial_{\Delta} f_{\text{all}}(s, x, \Delta) dx,$$

where

$$f_{\text{all}}(s, x, \Delta) = c'_v(1 - s) - t(x + \Delta) \quad \text{and hence} \quad \partial_{\Delta} f_{\text{all}}(s, x, \Delta) = -t.$$

Thus,

$$\frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{all}}(s, \Delta)} (-t) dx = -\frac{t}{\pi} |\mathcal{X}_{\text{all}}(s, \Delta)|.$$

(When summed with the adjacent slice, moving-boundary terms cancel at the switch point.)

A.5.5 Proof of Lemma 5.

Consider the slice where $\theta_r^*(s, x, \Delta) = \theta_{r,\text{int}}^*$ and the acceptance region lies fully within the posterior, i.e., $0 \leq x \leq \Delta$ and $\theta_{r,\text{int}}^* \leq \Delta - x$. From Lemma 3,

$$\theta_{r,\text{int}}^* = \frac{V - c_v(1 - s) - mc}{2ts}, \quad L(x, \Delta, \theta_{r,\text{int}}^*) = 2\theta_{r,\text{int}}^*.$$

The per- x first-order-condition integrand on this slice is therefore

$$f_{\text{int}}(s, x, \Delta) = \frac{L}{2\Delta} \left(c'_v(1 - s) - t\theta_{r,\text{int}}^* \right) = \frac{\theta_{r,\text{int}}^*}{\Delta} \left(c'_v(1 - s) - t\theta_{r,\text{int}}^* \right).$$

Let $\mathcal{X}_{\text{int}}(s, \Delta) \subseteq [0, \pi]$ denote the set of x where this regime is optimal. The contribution of this slice to the cross-partial is

$$\frac{\partial^2 \Pi_{\text{int}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{int}}(s, \Delta)} \partial_{\Delta} f_{\text{int}}(s, x, \Delta) dx.$$

Since $\theta_{r, \text{int}}^*$ and $c'_v(1 - s)$ are independent of Δ ,

$$\partial_{\Delta} f_{\text{int}}(s, x, \Delta) = -\frac{\theta_{r, \text{int}}^*}{\Delta^2} \left(c'_v(1 - s) - t \theta_{r, \text{int}}^* \right),$$

and thus

$$\frac{\partial^2 \Pi_{\text{int}}(s, \Delta)}{\partial s \partial \Delta} = -\frac{1}{\pi} \int_{\mathcal{X}_{\text{int}}(s, \Delta)} \frac{\theta_{r, \text{int}}^*}{\Delta^2} \left(c'_v(1 - s) - t \theta_{r, \text{int}}^* \right) dx.$$

Any moving-boundary terms at the edges of $\mathcal{X}_{\text{int}}(s, \Delta)$ cancel against adjacent regimes when the full cross-partial is assembled across all slices.

A.5.6 Proof of Lemma 6.

Consider signals x for which the posterior and the acceptance region partially overlap. From Lemma 3, the interior stationary point satisfies

$$\theta_{r, \text{par}}^*(s, x, \Delta) = \frac{V - c_v(1 - s) - mc}{2ts} - \frac{\Delta - x}{2},$$

which is feasible whenever $\theta_{r, \text{par}}^*$ lies strictly between the overlap bounds and below the feasible reach $\theta_{r, \text{feas}} = \frac{V - c_v(1 - s) - mc}{ts}$.

Let $\mathcal{X}_{\text{par}}(s, \Delta) \subseteq [0, \pi]$ denote the set of x where this regime is optimal. On this slice, the per- x first-order-condition integrand is

$$f_{\text{par}}(s, x, \Delta) = \frac{L(x, \Delta, r_{\text{par}}^*)}{2\Delta} (c'_v(1 - s) - t \theta_{r, \text{par}}^*) = \frac{\theta_{r, \text{par}}^* + \Delta - x}{2\Delta} (c'_v(1 - s) - t \theta_{r, \text{par}}^*).$$

Differentiating with respect to Δ gives

$$\begin{aligned} \partial_{\Delta} f_{\text{par}}(s, x, \Delta) &= \frac{x - \theta_{r, \text{par}}^*}{2\Delta^2} (c'_v(1 - s) - t \theta_{r, \text{par}}^*) + \frac{(\theta_{r, \text{par}}^* + \Delta - x) t}{4\Delta} \\ &= \frac{(c'_v(1 - s) - t \theta_{r, \text{par}}^*)}{4\Delta^2} \left(x - \frac{V - c_v(1 - s) - mc}{ts} \right) + \frac{L(x, \Delta, \theta_{r, \text{par}}^*) t}{4\Delta}. \end{aligned}$$

Hence the contribution of this slice to the cross-partial is

$$\frac{\partial^2 \Pi_{\text{par}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{par}}(s, \Delta)} \left[\frac{(c'_v(1-s) - t\theta_{r,\text{par}}^*)}{4\Delta^2} \left(x - \frac{V - c_v(1-s) - mc}{ts} \right) + \frac{L(x, \Delta, \theta_{r,\text{par}}^*) t}{4\Delta} \right] dx.$$

(*Moving boundary.*) The endpoints of $\mathcal{X}_{\text{par}}(s, \Delta)$ depend on Δ through feasibility conditions. If one wishes to compute the stand-alone derivative of this slice, the usual Leibniz boundary terms must be included. In the total decomposition

$$\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta} = \sum_R \frac{\partial^2 \Pi_R(s, \Delta)}{\partial s \partial \Delta},$$

these boundary terms cancel at regime switches since $\{\mathcal{X}_R\}_R$ partition $[0, \pi]$.

A.5.7 Proof of Proposition 4.

Write $a := \frac{V - c_v(1-s) - mc}{ts}$ and $c' := c'_v(1-s)$. Define

$$x_{\text{int}} := \Delta - \frac{a}{2}, \quad x_{\text{sw}} := a - 3\Delta, \quad x_{\text{max}} := a + \Delta.$$

For $\Delta \downarrow 0$ and $a > 0$, we have $x_{\text{int}} < 0$, so the acceptance–inside–posterior block is empty. Thus, only the sell–to–all and partial–overlap regimes are relevant.

- *Sell-to-all* ($\theta_r^* = x + \Delta$). The maximizing sell-to-all x -block is

$$\mathcal{X}_{\text{all}}(s, \Delta) = [\max\{0, x_{\text{int}}\}, \min\{\pi, x_{\text{sw}}\}]_+.$$

On this block $f_{\text{all}} = c' - t(x + \Delta)$, so $\partial_{\Delta} f_{\text{all}} = -t$. Hence the slice contribution is

$$\frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{all}}} (-t) dx = -\frac{t}{\pi} [\min\{\pi, x_{\text{sw}}\} - \max\{0, x_{\text{int}}\}]_+.$$

For small Δ (so $x_{\text{int}} < 0$), this simplifies to

$$\frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} = -\frac{t}{\pi} [\min\{\pi, a - 3\Delta\}]_+.$$

- *Partial overlap* ($\theta_r^* = \theta_{r,\text{par}}^*$). Here

$$\theta_{r,\text{par}}^* = \frac{a - \Delta + x}{2}, \quad L = \theta_{r,\text{par}}^* + \Delta - x = \frac{a + \Delta - x}{2}.$$

Differentiating the per- x integrand gives

$$\partial_{\Delta} f_{\text{par}}(x, \Delta) = \frac{(x - a)(c' - t\theta_{r,\text{par}}^*)}{4\Delta^2} + \frac{t}{4\Delta} L(x, \Delta, \theta_{r,\text{par}}^*).$$

Let $u := x - a$. Then on the maximizing partial block

$$\mathcal{X}_{\text{par}}(s, \Delta) = (\max\{0, x_{\text{sw}}\}, \min\{\pi, x_{\text{max}}\}),$$

and when there is no clipping (i.e., $a < \pi$ and Δ small) we have $u \in (-3\Delta, \Delta)$ and

$$\partial_{\Delta} f_{\text{par}}(u, \Delta) = \frac{t}{8} + \frac{(c' - ta)u}{4\Delta^2} - \frac{t u^2}{8\Delta^2}.$$

Integrating over $u \in (-3\Delta, \Delta)$ yields

$$\frac{\partial^2 \Pi_{\text{par}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{par}}} \partial_{\Delta} f_{\text{par}} dx = \frac{1}{\pi} \left[-(c' - ta) - \frac{2}{3} t \Delta \right].$$

(If a is near 0 or π , clip the limits; this does not affect the 0^+ limit below.)

If $a \leq \pi$, summing the two contributions gives

$$\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta} = \frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} + \frac{\partial^2 \Pi_{\text{par}}(s, \Delta)}{\partial s \partial \Delta} = -\frac{t}{\pi} x_{\text{sw}} + \frac{1}{\pi} \left[-(c' - ta) - \frac{2t\Delta}{3} \right] \xrightarrow{\Delta \rightarrow 0^+} -\frac{c'_v(1-s)}{\pi} < 0.$$

If $a > \pi$, then for sufficiently small $\Delta < \frac{a-\pi}{3}$ (which holds as $\Delta \downarrow 0$), the entire domain is sell-to-all, giving

$$\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta} = \frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} = -t < 0.$$

Thus, in all cases,

$$\frac{\partial^2 \Pi(s^*(0), 0^+)}{\partial s \partial \Delta} < 0, \quad \text{and since} \quad \frac{\partial^2 \Pi(s^*(0), 0^+)}{\partial s^2} < 0,$$

it follows that

$$\frac{ds^*}{d\Delta}(0^+) = - \frac{\frac{\partial^2 \Pi}{\partial s \partial \Delta}}{\frac{\partial^2 \Pi}{\partial s^2}} \Big|_{(s^*(0), 0^+)} < 0.$$

A.5.8 Proof of **Proposition 5**.

Fix (s, Δ) and write

$$a := \frac{V - c_v(1-s) - mc}{ts} = \theta_{r,\text{feas}}(s), \quad c' := c'_v(1-s).$$

Assume $\Delta < \frac{a}{3}$. This ensures both that the acceptance-inside-posterior regime is empty (which alone requires only $\Delta < \frac{a}{2}$, since $\theta_{r,\text{int}}^* = \frac{a}{2}$ cannot fit inside the posterior when $\Delta < \frac{a}{2}$) and that the sell-to-all window $[0, a - 3\Delta]$ is nonempty (which requires $\Delta < \frac{a}{3}$). Since $\frac{a}{3} < \frac{a}{2}$, the latter is the binding requirement, and only the sell-to-all and partial-overlap slices are relevant on $x \in [0, \pi]$.

Sell-to-all. Here $\theta_r^* = x + \Delta$ and the per- x FOC integrand is $f_{\text{all}}(s, x, \Delta) = c' - t(x + \Delta)$, so $\partial_\Delta f_{\text{all}} = -t$. The optimal window is $\mathcal{X}_{\text{all}}(s, \Delta) = [0, a - 3\Delta]$. Hence

$$\frac{\partial^2 \Pi_{\text{all}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \int_{\mathcal{X}_{\text{all}}} (-t) dx = -\frac{t}{\pi}(a - 3\Delta).$$

Partial overlap. Here $\theta_{r,\text{par}}^* = \frac{a - \Delta + x}{2}$ and $L(x, \Delta, \theta_{r,\text{par}}^*) = \frac{a + \Delta - x}{2}$. The per- x FOC integrand is

$$f_{\text{par}}(s, x, \Delta) = \frac{L}{2\Delta}(c' - t\theta_{r,\text{par}}^*) = \frac{\theta_{r,\text{par}}^* + \Delta - x}{2\Delta}(c' - t\theta_{r,\text{par}}^*),$$

so

$$\partial_\Delta f_{\text{par}}(s, x, \Delta) = \frac{(x - a)(c' - t\theta_{r,\text{par}}^*)}{4\Delta^2} + \frac{t}{4\Delta} L(x, \Delta, \theta_{r,\text{par}}^*).$$

On the optimal window $\mathcal{X}_{\text{par}}(s, \Delta) = (a - 3\Delta, a + \Delta)$ let $u := x - a \in (-3\Delta, \Delta)$. Then

$$\partial_\Delta f_{\text{par}}(u, \Delta) = \frac{t}{8} + \frac{(c' - ta)u}{4\Delta^2} - \frac{tu^2}{8\Delta^2}.$$

Integrating u over $(-3\Delta, \Delta)$ gives

$$\frac{\partial^2 \Pi_{\text{par}}(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \left[-(c' - ta) - \frac{2}{3}t\Delta \right].$$

Summing slices. Adding the two contributions,

$$\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta} = \frac{\partial^2 \Pi_{\text{all}}}{\partial s \partial \Delta} + \frac{\partial^2 \Pi_{\text{par}}}{\partial s \partial \Delta} = \frac{1}{\pi} \left[-c' + \frac{7}{3}t\Delta \right].$$

(Leibniz moving-boundary terms at the switching points cancel between adjacent slices because the windows partition $[0, \pi]$.)

Thus, under $\Delta < \frac{a}{2} = \frac{\theta_{r,\text{feas}}(s)}{2}$,

$$\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta} = \frac{1}{\pi} \left[-c'_v(1 - s) + \frac{7}{3}t\Delta \right],$$

which is positive iff $\Delta > \frac{3}{7t} c'_v(1 - s)$. Evaluating this identity at $s = s(\Delta)$ yields the statement of the proposition.

A.5.9 Proof of Corollary 1.

Under $c_v(1-s) = (1-s)^2$, we have $c'_v(1-s) = 2(1-s)$. Proposition 5 states that

$$\frac{\partial^2 \Pi(s, \Delta)}{\partial s \partial \Delta} > 0 \iff \Delta > \frac{3}{7t} c'_v(1-s).$$

Hence, for any $s \in [s_{\min}, 1]$,

$$\Delta > \frac{3}{7t} c'_v(1-s) = \frac{6}{7t}(1-s) \Rightarrow \Delta > \frac{6}{7t}(1-s_{\min}),$$

which yields the uniform lower bound.

The intermediate- Δ region additionally requires $\Delta < \frac{\theta_{r,\text{feas}}(s)}{2}$ for all s , where

$$\theta_{r,\text{feas}}(s) = \frac{V - mc - c_v(1-s)}{ts} = \frac{V - mc - (1-s)^2}{ts}.$$

Thus a uniform upper bound is

$$\bar{\Delta} = \inf_s \frac{\theta_{r,\text{feas}}(s)}{3} = \min \left\{ \frac{V - mc}{3t}, \frac{V - mc - (1-s_{\min})^2}{3ts_{\min}} \right\}.$$

Imposing $\Delta \in [0, \pi]$ gives the stated interval:

$$\frac{6}{7t}(1-s_{\min}) < \Delta < \bar{\Delta}.$$

A.5.10 Proof of Lemma 7

Given s, p , consumer located at $\theta_c = \theta$ has utility

$$u(\theta) = V - c_v(1-s) - ts\theta - p_{\text{all}} = ts(x + \Delta - \theta).$$

Thus, consumer surplus is given as

$$\begin{aligned} CS(s, x, \Delta) &= \frac{1}{2\Delta} \int_{x-\Delta}^{x+\Delta} ts(x + \Delta - \theta) d\theta \\ &= ts\Delta. \end{aligned}$$

A.5.11 Proof of Lemma 8

Given s, p , consumer located at $\theta_c = \theta$ has utility

$$u(\theta) = V - c_v(1 - s) - ts\theta - p_{\text{int}}^* = ts(\theta_{r,\text{int}}^* - |\theta|),$$

where $p_{\text{int}}^* = V - c_v(1 - s) - ts\theta_{r,\text{int}}^*$. Thus, consumer surplus is given as

$$\begin{aligned} CS(s, x, \Delta) &= \frac{1}{2\Delta} \int_{-\theta_{r,\text{int}}^*}^{\theta_{r,\text{int}}^*} ts(\theta_{r,\text{int}}^* - |\theta|) d\theta = \frac{2ts}{2\Delta} \int_0^{\theta_{r,\text{int}}^*} (\theta_{r,\text{int}}^* - \theta) d\theta \\ &= \frac{(V - c_v(1 - s) - m)^2}{8ts\Delta}. \end{aligned}$$

A.5.12 Proof of Lemma 9

Given s, p , consumer located at $\theta_c = \theta$ has utility

$$u(\theta) = V - c_v(1 - s) - ts\theta - p_{\text{par}}^* = ts(\theta_{r,\text{par}}^* - \theta),$$

where $p_{\text{par}}^* = V - c_v(1 - s) - ts\theta_{r,\text{par}}^*$. Thus, consumer surplus is given as

$$\begin{aligned} CS(s, x, \Delta) &= \frac{1}{2\Delta} \int_{x-\Delta}^{\theta_{r,\text{par}}^*} ts(\theta_{r,\text{par}}^* - \theta) d\theta \\ &= \frac{ts}{4\Delta} \left(\frac{V - c_v(1 - s) - mc}{2ts} + \frac{\Delta - x}{2} \right)^2. \end{aligned}$$

A.5.13 Proof of Proposition 6.

The signs follow from the regime-specific derivatives stated in the proposition. For partial overlap, feasibility implies that $x \in (a - \Delta, a + \Delta)$, where $a = \frac{V - c_v(1 - s) - mc}{ts}$. Thus, it follows that $a + \Delta - x > 0$ and $\Delta - a + x > 0$, proving that the derivative is positive in this case.

A.5.14 Proof of Proposition 7.

(Direct effect.) For fixed s , as $\Delta \downarrow 0$ the partial overlap window has x -measure $O(\Delta)$ and contributes $O(\Delta^2)$ to CS , while the sell-to-all slice covers the remaining mass and yields $CS_{\text{all}} = ts\Delta$. Hence

$$CS(s, \Delta) = ts\Delta + O(\Delta^2) \quad \Rightarrow \quad \partial_{\Delta} CS(s, \Delta) = ts + O(\Delta).$$

Evaluating at $s = s^*(0)$ gives $\partial_{\Delta} CS(s^*(0), 0^+) = ts^*(0) > 0$.

(*Design effect.*) For fixed Δ , on sell-to-all we have $CS_{\text{all}} = ts\Delta$, so $\partial_s CS_{\text{all}} = t\Delta$; the partial overlap window has measure $O(\Delta)$ and integrand $O(ts\Delta)$, hence it contributes $O(\Delta)$ to $\partial_s CS$ as well. Thus

$$\partial_s CS(s, \Delta) = O(\Delta) \quad (\Delta \downarrow 0).$$

(*Along s^* and the chain rule.*) Using

$$\frac{d}{d\Delta} CS(s^*(\Delta), \Delta) = \partial_\Delta CS(s^*, \Delta) + \partial_s CS(s^*, \Delta) \frac{ds^*}{d\Delta},$$

we note that, even though $\frac{ds^*}{d\Delta}(0^+) < 0$, the product $\partial_s CS(s^*, \Delta) \cdot \frac{ds^*}{d\Delta} = O(\Delta) \rightarrow 0$ as $\Delta \downarrow 0$, because $\partial_s CS = O(\Delta)$. Therefore the limit of the total derivative at 0^+ equals the direct term:

$$\left. \frac{d}{d\Delta} CS(s^*(\Delta), \Delta) \right|_{0^+} = \lim_{\Delta \downarrow 0} \partial_\Delta CS(s^*, \Delta) = ts^*(0) > 0.$$

A.6 Numerical Example.

A.6.1 Weak Privacy Regulation

Assume that there is weak privacy regulation. If the seller only adjusts prices (holding s fixed), the seller must shift the price schedule down for all signals, seen in the left panel of [Figure 9](#). Instead, if the seller adjusts product design, there is a rotation in the price schedule, as seen in the right panel of [Figure 9](#).

The resulting profit rotation indicates that the seller's optimal product design should become more broad with this increase in privacy.

A.6.2 Intermediate Privacy Regulation

Now assume that there is intermediate privacy regulation. If the seller only adjusts prices (holding s fixed), the seller must increase the price for some consumers, and decrease for others, seen in [Figure 10](#). Instead, if the seller adjusts product design, the price and design become strategic complements, moving in the same direction, as seen in the right panel of [Figure 10](#).

The resulting profit rotation indicates that choosing a more niche product design is optimal for the seller in response to an increase in privacy.

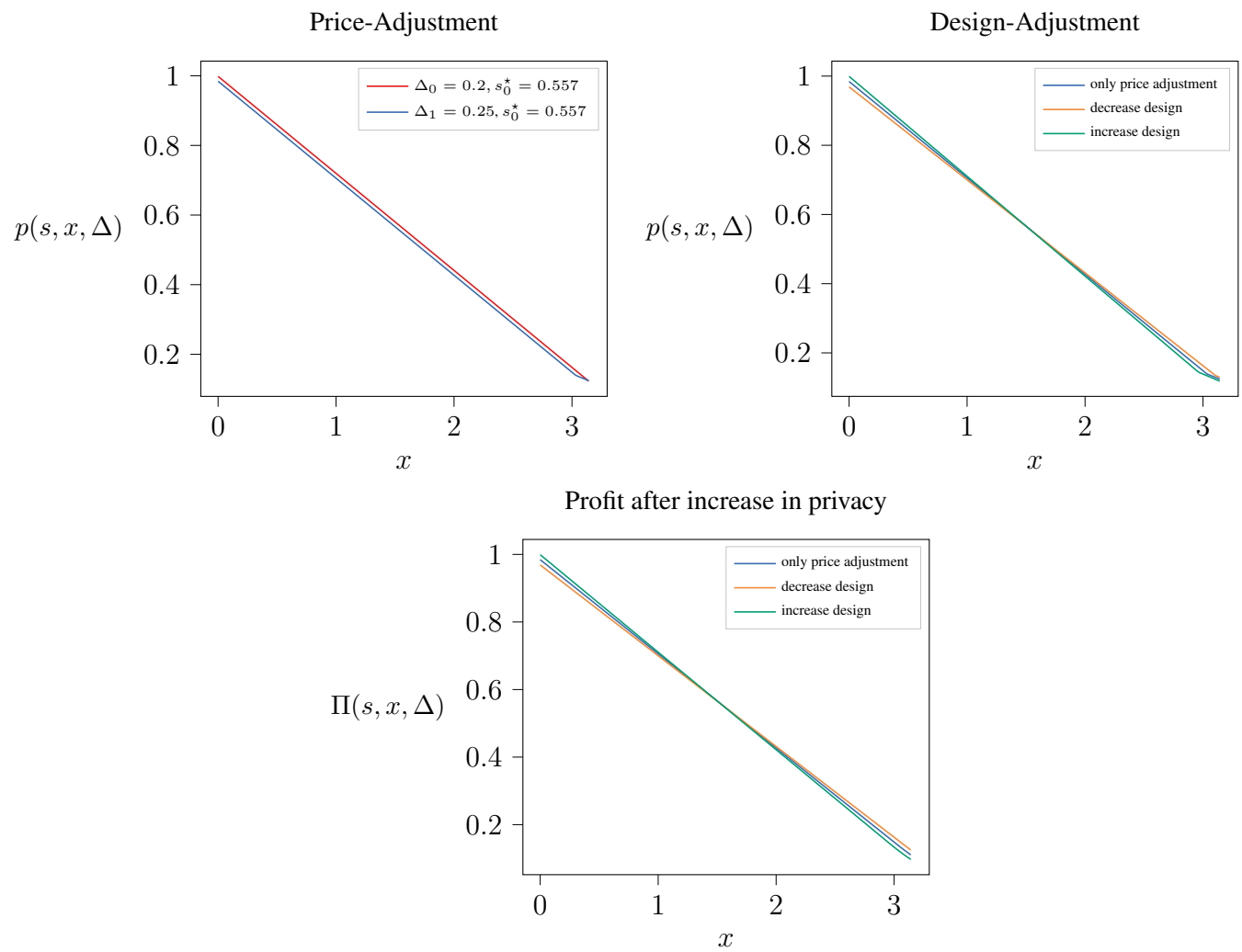


Figure 9: *Small Δ :* $\mathbb{E}[\Pi(\Delta_1, s \text{ fixed})] = 0.5467$
 $\mathbb{E}[\Pi(\Delta_1, s \downarrow)] = 0.5468$
 $\mathbb{E}[\Pi(\Delta_1, s \uparrow)] = 0.5458$

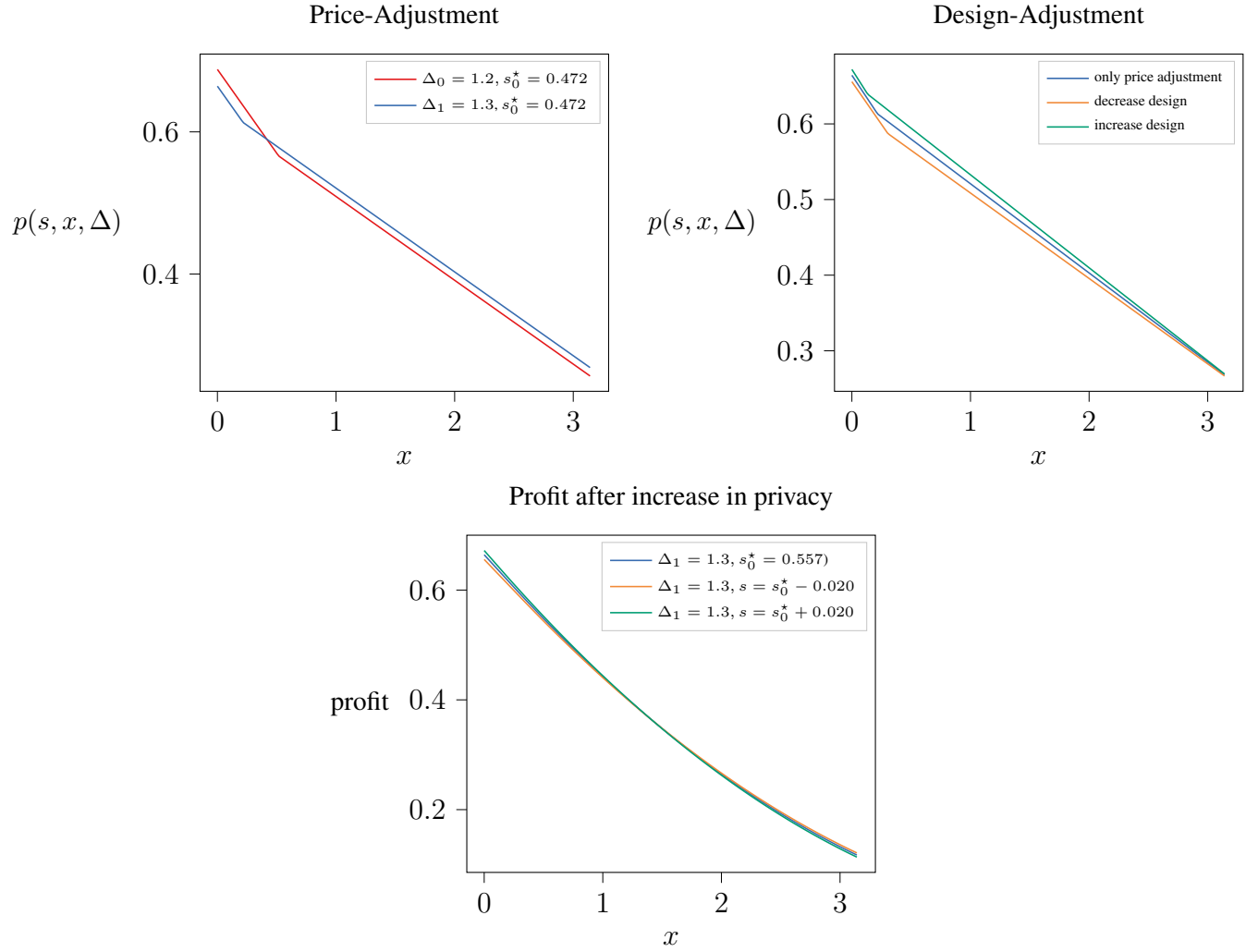


Figure 10: *Intermediate Δ exercise.*

Intermediate Δ : $\mathbb{E}[\Pi(\Delta_1, s \text{ fixed})] = 0.3540$
 $\mathbb{E}[\Pi(\Delta_1, s \downarrow)] = 0.3537$
 $\mathbb{E}[\Pi(\Delta_1, s \uparrow)] = 0.3541$